A Survey of Threshold Signatures: NIST Standards, Post-Quantum Cryptography, Exotic Techniques, and Real-World Applications

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Threshold digital signatures enable a distributed execution of signature functionalities and will play a crucial role in the security of emerging decentralized next-generation networked systems and applications. In this article, we provide a comprehensive and systematic survey of threshold and distributed signatures with advanced features. Our survey encompasses threshold signatures in conventional and post-quantum cryptography (PQC) settings and captures custom-design and standard signatures (e.g., conventional NIST and NIST-PQC). We examine both generic (via secure multi-party computation) and custom thresholding techniques for a myriad of signature families while investigating exotic signatures, real-life applications, and potential future research directions.

CCS Concepts: • **Security and privacy → Digital signatures;**

Additional Key Words and Phrases: Threshold signatures, distributed signatures, NIST standards, post-quantum cryptography, exotic signatures, secure multi-party computation

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1 Introduction

In the digital age, ensuring the integrity, authenticity, and non-repudiation of information is critical. Digital signatures, enabled by **Public Key Infrastructure** (**PKI**) [156], play a central role in achieving these objectives. To promote standardization, the **National Institute of Standards and Technology** (**NIST**) [79] has defined a suite of approved algorithms, including ECDSA [9], EdDSA [34, 174], RSA [173, 226], and pairing-based schemes. Moreover, notable efficient implementations of **Discrete Logarithm Problem** (**DLP**)-based schemes such as Diffie-Hellman and

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Schnorr-like signatures [262] on **Elliptic Curves** (**ECs**) have been proposed [34, 78, 86] for deployment in applications like the automotive supply chain, IoT [236, 293], and wireless networks [154]. Each scheme offers distinct tradeoffs, enabling application-specific security optimization. However, these conventional digital signatures face significant challenges in emerging **Next-Generation** (**NextG**) distributed systems, particularly under **post-quantum** (**PQ**) threats [216, 253, 295].

Quantum computing poses a significant threat to sectors such as finance, IoT, government, and defense, as it can break widely deployed cryptographic primitives, particularly digital signatures based on **elliptic curve and discrete logarithm problems** (**EC/DLPs**). Shor's algorithm [265] enables efficient quantum attacks on these assumptions via quantum parallelism and superposition. This risk is amplified by the accelerating quantum industry, with IDC projecting a $8.6 billion market by 2027. Advancements by major players, Google's Bristlecone, China's Zuchongzhi and Jiuzhang 2.0, and IBM's Osprey, highlight the urgency for PQ security. To address this, the field of **Post-Quantum Cryptography** (**PQC**) has emerged, focusing on developing cryptographic algorithms resilient against both classical and quantum adversaries. NIST initiated a PQC standardization process in 2017 [99, 234], culminating in 2022 by selection of four algorithms [235]: ML-KEM [252] for **public-key encryption** (**PKE**) and key establishment, and ML-DSA [238], FALCON [135], and SLH-DSA [237] for digital signatures. Although NIST has adopted these as standards, PQC remains an active research area, with NIST's second round of standardization aimed at diversifying the hardness assumptions for signatures [1].

A fundamental limitation of both classical and PQ-secure digital signatures is their reliance on centralized trust, where a single party manages cryptographic secrets and operations. This centralization creates a single point of failure. *Threshold cryptography* mitigates this risk by distributing secrets (e.g., via Shamir's secret sharing [263]) and computations using **Multi-Party Computation** (**MPC**) [98], requiring collaboration among a subset of parties to perform cryptographic operations. Recognizing its benefits in enhancing security, fault tolerance, and robustness, NIST launched an initiative [55] to standardize multi-party threshold schemes. This effort evaluates submissions for best practices, reusable components, and applicability to domains such as cryptocurrencies [142], blockchain [242], and IoT [192]. Submissions fall into two categories: (i) threshold schemes built on NIST-approved primitives and (ii) those based on alternative algorithms with comparable or extended functionality. Given the critical need for PQ resilience, developing threshold variants of PQ-safe schemes is a priority. Cozzo et al. [87] has offered a detailed analysis of threshold adaptations for second-round NIST PQC candidates, highlighting design strategies and associated computational costs.

In addition to regular digital signatures, various forms of distributed signatures with specialized properties, often referred to as *exotic signatures*, have been proposed. These include, but are not limited to, Group signatures [76], where a group member can anonymously sign on behalf of the group with traceability; Ring signatures [254], which enable a signer to anonymously generate a signature within an arbitrary set of users; and Multi-signatures [170], where multiple parties collaboratively generate a single, compact signature. These schemes are extensively used in privacy-sensitive domains such as blockchain [42], cryptocurrencies [179], IoT [197], and cloud computing [280]. Beyond offering advanced privacy properties, group and ring signatures can also be extended with threshold properties for enhanced robustness, while multi-signatures inherently resemble threshold schemes via participation from all signers.

In conclusion, threshold cryptography, particularly those with exotic properties, is vital for real-world applications, boosting security, resilience, and robustness through privacy-preserving collaborative computation. The following section reviews the current state-of-the-art literature on these signature schemes, emphasizing which surveys address the challenges we discussed and how thoroughly they cover all relevant aspects.

Table 1. Summary of Published Surveys on Threshold and Exotic Signatures

Work by	MPC[1]	Conventional Signatures[2]					PQ Security[3]		Exotic (special) Signatures[4]			Applications
		Schnorr	ECDSA	EdDSA	RSA	Pairings	NIST	Non-NIST	Multi	Group	Ring	
Liu et al. (2024) [205]	✗	✗	✗	✗	✗	✗	✓	✓	✗	✓	✓	✗
Buser et al. (2023) [59]	✗	✗	✗	✗	✗	✗	✗	✓	✓	✗	✓	Blockchain
Csahin et al. (2023) [259]	✗	✗	✗	✗	✗	✗	✗	✓	✗	✓	✗	✗
Ullah et al. (2023) [275]	✗	✗	✓	✗	✗	✗	✗	✗	✗	✗	✗	✗
Perera et al. (2022) [244]	✗	✗	✗	✗	✗	✗	✗	✗	✗	✓	✓	E-voting, E-commerce
Feng et al. (2022) [129]	✓	✗	✗	✗	✗	✗	✗	✓	✗	✓	✗	✗
Zhang et al. 2021 [299]	✗	✗	✗	✗	✗	✗	✗	✓	✗	✓	✗	✗
Orsini (2021) [241]	✓	✗	✗	✗	✗	✗	✗	✗	✗	✗	✗	✗
Aumasson et al. (2020) [13]	✗	✗	✓	✗	✗	✗	✗	✗	✗	✗	✗	✗
Lindell (2020) [202]	✓	✗	✗	✗	✗	✗	✗	✗	✗	✗	✗	Privacy-preserving ML
Fang et al. (2020) [127]	✗	✗	✓	✗	✗	✗	✗	✓	✗	✗	✗	Blockchain
Ergezer et al. (2020) [120]	✗	✓	✓	✗	✗	✓	✗	✓	✗	✗	✗	Certificate Authorities, Bitcoin
Cozzo et al. (2019) [87]	✓	✗	✗	✗	✗	✗	✓	✗	✗	✗	✗	
Zhao et al. (2019) [302]	✓	✗	✗	✗	✗	✗	✗	✗	✗	✗	✗	ML, Private set operations
Al-Zubaidie et al. (2019) [7]	✗	✗	✓	✗	✗	✗	✗	✗	✗	✗	✗	E-commerce
Al-Kaabi et al. (2019) [6]	✗	✗	✗	✗	✓	✗	✗	✗	✗	✗	✗	✗
Ours	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	Blockchain, Cryptocurrency, IoT, Cloud, Healthcare, Network Security, E-governance

[1]Generic thresholding without standard deviation using secure computation; [2]Conventional signatures standardized by NIST and approaches used for constructing threshold schemes upon them; [3]NIST PQC: thresholding standardized post-quantum signatures; Non-std PQC: thresholding non-standard PQ-secure schemes; [4]Exotic signatures and covering their threshold construction.

1.1 The State-of-the-art Surveys and Limitations

We have conducted a comprehensive and systematic investigation of recent studies focusing on threshold conventional and PQ-secure digital signatures, as well as possible threshold constructions of signatures with exotic properties. The findings of our analysis are summarized in Table 1. In our examination, we considered various aspects of these studies, including the **Approaches** adopted (generic (transparent) using MPC or non-generic) (Column 2), the **Schemes** that were studied including conventional ones (Column 3) or NIST standardized and non-standard PQ-secure signatures (Column 4), the **Special schemes** studied including the group, ring, and multi-signatures (Column 5), and the **Application** areas covered for the use of the studied signature(s) (Column 6).

An initial observation from Table 1 reveals a significant gap in existing literature concerning surveys that address NIST PQC standards and exotic signatures through the lens of threshold cryptography. *(i)* Current surveys discuss distributed signatures primarily from the perspective of custom thresholding, with limited coverage of generic (MPC-based) approaches, and lack a comparative analysis across both. *(ii)* This gap is exacerbated by the absence of unified treatment of threshold adaptations, custom and MPC-based, of conventional and NIST-PQC schemes, as well as a broad class of exotic signatures (e.g., [76, 170, 254]). The need for such coverage is underscored by NIST's recent **Multi-Party Threshold Cryptography** (**MPTC**) initiative [55]. *(iii)* Finally, existing surveys have a restricted application domain focus (e.g., blockchain), overlooking broader real-world deployment contexts. We elaborate on these gaps below:

Limited Coverage on Thresholding Approaches and Their Comparison: The custom-design of distributed signatures such as multi-signatures [206], group signatures [49], and some custom-design threshold signatures [188] received attention in the literature. Yet another angle to threshold digital signature is the use of MPC [87]. Although MPC and some of its applications have been surveyed in the literature [129, 202, 241, 302], only little attention (e.g., [59]) has been given to transparent thresholding of signatures with MPC [55, 87]. In particular, there is a gap in encompassing transparent thresholding of digital signatures, possibly with exotic properties (whether conventional or PQ-safe).

Narrow/Selective Focus on Conventional-Custom and Exotic Distributed Signatures: Existing surveys predominantly focus on selected groups of signatures, such as EC-based schemes [275] and ECDSA [7, 13], while overlooking others of equal relevance, like threshold BLS with applications in IoT [155] and blockchain [260], or threshold Schnorr in cryptocurrency settings [231, 232]. Similarly, although valuable surveys on exotic signatures exist [59, 129, 205, 244, 259, 299], they often focus on specific application domains (e.g., blockchain, IoT) and treat exotic features in isolation. The lack

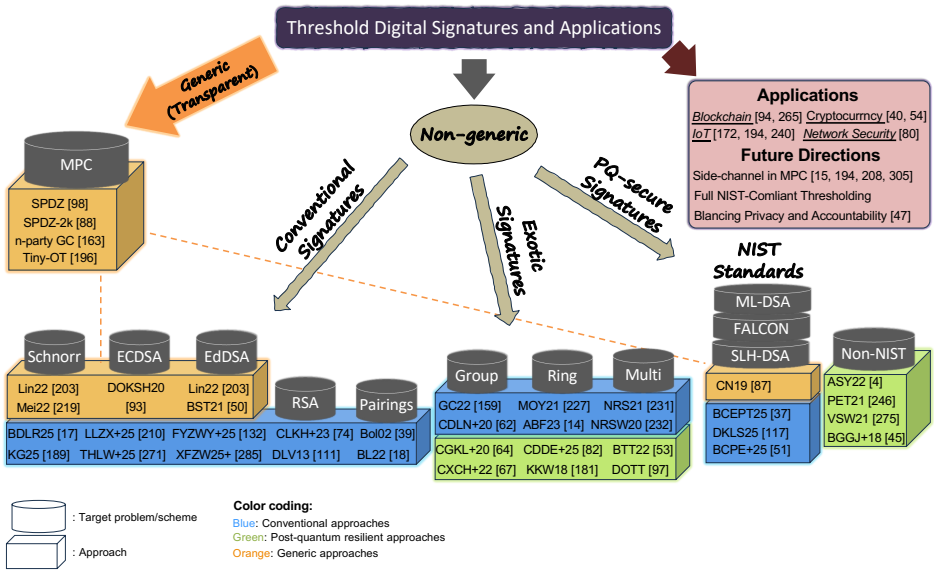


Fig. 1. High-level taxonomy of our survey.

of discussion on threshold constructions of exotic signatures in the state-of-the-art may reinforce the misconception that threshold signatures are confined to classical schemes (e.g., ECDSA and RSA) or PQ-secure ones like ML-DSA. In contrast, in reality, thresholding can be integrated with the special properties offered by these signatures. Moreover, threshold adaptations of PQ digital signatures, including both NIST-PQC standardized schemes and custom designs [50], remain insufficiently explored. While some efforts have investigated non-standard thresholding in the context of blockchain [59] or proposed threshold variants of NIST-PQC candidates [87], a comprehensive survey of standardized, custom, and exotic threshold signatures is lacking, especially through the lens of secure MPC (as highlighted in Table 1).

Limited Attention to Broad Applications and Future Directions: The practical applications of threshold signatures have significant importance, yet we observed that only a limited number of articles have examined applications of threshold signatures, primarily centered around Blockchain (e.g., [59]) and e-voting/currency (e.g., with PQC [259]) domains. Furthermore, it is ideal to consider visions towards the future evolution of threshold digital signatures. However, we observe that this important angle received less attention in the majority of the current surveys. Hence, there is a need for forward-looking perspectives in the design and applications of emerging threshold digital signatures.

1.2 Our Contribution

This article addresses key gaps in the literature by presenting the first comprehensive survey of threshold digital signatures spanning both conventional and PQC settings (see Figure 1). It covers NIST PQC standards and custom constructions using both generic (e.g., MPC-based) for transparency and non-generic thresholding techniques. Beyond these well-known signatures, our survey seeks to highlight that threshold constructions extend beyond classical or PQ-secure signatures and can also be integrated with signatures featuring special properties, making them applicable to privacy-sensitive applications. Figure 1 shows the taxonomy of our survey, and our main contributions are as follows:

Extensive Coverage of Custom and Generic Thresholding with a Vast Range of Digital Signatures: This survey investigates both custom-designed and generic thresholding approaches across a wide range of digital signature families to provide comprehensive coverage of distributed digital signatures. Emphasizing MPC-based thresholding, and aligned with NIST's recent initiatives [55], we argue that such generic mechanisms will be integral to future threshold signature standards. MPC-based thresholding enables *transparent thresholding*, preserving the algorithmic structure, provable security, signature size, and verification properties of the original scheme, crucial for compliance-driven real-world deployments (e.g., government). Despite its merits, generic thresholding is computationally costlier than custom designs, which dominate the landscape. In Section 6, we introduce generic MPC-based thresholding and review its application to widely-used schemes like MPC-Schnorr [219], MPC-ECDSA [9], and MPC-EdDSA [50], further extended to NIST-PQC signatures in Section 7. We also present relevant MPC gadgets and highlight the connections between custom and generic thresholding, particularly in the context of PQC, with detailed comparisons summarized in our tables.

Comprehensive Examination of Threshold Conventional and PQ-Secure Signatures: This survey presents a detailed analysis of threshold digital signatures across major cryptographic families, including those based on the *DLP*, *ECDLP*, *Cryptographic Pairing*, ***Integer Factorization (IF)***, and *PQ* assumptions. (i) Section 3.1 explores Schnorr-type signatures and standardized schemes such as ECDSA and EdDSA, highlighting their threshold adaptations, practical relevance, and performance tradeoffs. (ii) Sections 4 and 5 cover pairing-based (BLS) and IF-based (RSA) threshold schemes, respectively. (iii) Section 7 addresses threshold PQ-secure signatures, encompassing both custom and MPC-based designs. (iv) Each section includes structured, chronological tables summarizing key works, core techniques, and, if applicable, performance metrics to enable comparative evaluation.

Investigation of Exotic Features, Real-life Applications, and Future Directions: In the context of signatures with exotic features, Section 8 first provides a succinct analysis of the main constructions for each scheme, followed by a discussion of recent efforts to develop their threshold variants for enhanced robustness. We focus on Group [303] (Section 8.1), Ring [30] (Section 8.2), and Multi-signatures [231] (Section 8.3) in conventional and PQ-secure settings. In Section 9, we examine key real-world applications of threshold and exotic signature schemes in emerging domains. Finally, in Section 10, we outline current challenges and open problems in the design and implementation of threshold signatures (e.g., from the side-channel security perspective), and discuss future research directions.

Structure of the Survey: Section 2 provides background, definitions, and security models. Section 3 covers threshold constructions for Schnorr-like signatures, including ECDSA and EdDSA. Section 4 focuses on pairing-based schemes (e.g., BLS), and Section 5 examines threshold RSA. Section 6 discusses MPC-based thresholding [55], extended to PQ-secure schemes in Section 7. Section 8 surveys exotic signatures and their threshold/PQ variants. Section 9 outlines practical use cases, and Section 10 highlights open problems and future directions.

2 Preliminaries

Notations: H denotes a secure cryptographic hash function (e.g., SHA-256); $||.||$ is the infinity norm; $||$ denotes string concatenation. p and q are large primes; $m \in \{0,1\}^*$ is a message of arbitrary finite length. sk and PK represent private and public keys, respectively. G and g refer to a cyclic group and its generator. $CURV$, O, B, and n denote an EC, its identity element, base point, and order. $\times$ indicates scalar multiplication on the curve. Scalars are denoted in regular font x; vectors in bold $\mathbf{x}$. $\mathbb{Z}_p^*$ denotes a cyclic group of order $p-1$, and $\mathbb{F}_p$ the finite field of the same order.

Table 2. Cryptographic Abbreviations

Symbol	Description	Symbol	Description
IF	Integer Factorization	PQC	Post-quantum Cryptography
(EC)DLP	(Elliptic Curve) Discrete Logarithm Problem	LWE	Learning With Errors
(EC)DSA	(Elliptic Curve) Digital Signature Algorithm	SIS	Short Integer Solution
EdDSA	Edwards-curve DSA	(NI)ZKP	(Non-interactive) Zero-knowledge Proof
(C/D)DH	(Computational/Decisional) Diffie–Hellman	SS	Secret Sharing

2.1 Intractability Assumptions and Mathematical Backgrounds

We concisely overview the intractability assumptions and definitions used in this work (see abbreviations in Table 2).

Definition 2.1.1 (IF Problem). Given a composite integer $N = p \cdot q$, the problem is finding its prime divisors.

Definition 2.1.2 (DLP Problem). Given the group $\mathbb{Z}_p^*$ with generator g and an arbitrary $\beta \in \mathbb{Z}_p^*$, the problem is finding the integer x, such that $g^x \equiv \beta \bmod p$.

Definition 2.1.3 (ECDLP Problem). Given the elliptic curve $CURV$ and an arbitrary point $Q \in CURV$, the problem is finding the integer $1 \leq x \leq n-1$, such that $Q = x \times B$.

Definition 2.1.4 (DDH Problem). Given the group G with generator g and three elements of the group $a = g^x$, $b = g^y$, and $c = g^z$, the problem is to decide whether $g^z = g^{xy}$.

Definition 2.1.5 (Lattice). Let $\mathbb{R}^m$ be the m-dimensional Euclidean space, a lattice is a set of m-dimensional points constructed by a linear combination of a set of independent vectors known as basis $\mathbf{B} = (\mathbf{b}_1, \mathbf{b}_2, ..., \mathbf{b}_n)$ as follows:

$$\mathcal{L}(\mathbf{b}_1, \mathbf{b}_2, ..., \mathbf{b}_n) = \{\Sigma_{i=1}^n x_i \mathbf{b}_i \ : \ x_i \in \mathbb{Z} \ , \ m \geq n\}$$

Definition 2.1.6 (SIS Problem). Assume β is a positive real, and m is an integer. Given m uniformly random vectors $\mathbf{a}_i \in \mathbb{Z}_q^n$ forming the matrix $\mathbf{A} \in \mathbb{Z}_q^{n \times m}$, the problem is to find a nonzero vector $\mathbf{z} \in \mathbb{Z}^m$ ($||\mathbf{z}|| \leq \beta$) such that $\mathbf{Az} = \mathbf{0} \in \mathbb{Z}_q^n$.

Definition 2.1.7 (LWE Problem). Let $\mathbf{s} \in \mathbb{Z}_q^n$ be a secret vector and χ be an error distribution over $\mathbb{Z}$. The LWE distribution $\mathcal{A}_{\mathbf{s},\chi}^{LWE}$ over $\mathbb{Z}_q^n \times \mathbb{Z}_q$ is sampled by choosing $\mathbf{a} \xleftarrow{\$} \mathbb{Z}_q^n$ and $e \leftarrow$, and outputting $((\mathbf{a}, b) = \langle \mathbf{s}, \mathbf{a} \rangle + e \bmod q)$. The search LWE is to find $\mathbf{s}$ given m independent samples $(\mathbf{a}_i, b_i) \in \mathbb{Z}_q^n \times \mathbb{Z}_q$ drawn from the $\mathcal{A}_{\mathbf{s},\chi}^{LWE}$.

*Definition 2.1.8 (**Module LWE (MLWE)** Problem).* Let $\mathcal{R}$ be a ring with field of fraction of K, $M = \mathcal{R}^d$, $\mathbf{s} \in \mathbb{Z}_q^n$ be a secret vector and χ be an error distribution over $\mathcal{R}$. The M-LWE distribution $\mathcal{A}_{\mathbf{s},\chi}^{M-LWE}$ over $\mathcal{R}_q^d \times \mathcal{R}_q$ is sampled by choosing $\mathbf{a} \leftarrow \mathcal{R}_q^d$ uniformly at random, choosing $e \leftarrow \chi$, and outputting $((\mathbf{a}, b) = \langle \mathbf{s}, \mathbf{a} \rangle + e \bmod q)$. The decision M-LWE is to distinguish between m independent samples of $\mathcal{A}_{\mathbf{s},\chi}^{M-LWE}$ and uniform distribution on $\mathcal{R}_q^d \times \mathcal{R}_q$.

*Definition 2.1.9 (**Module SIS (MSIS)** Problem).* Assume β is a positive real. Given $\mathcal{R}$ the ring of integers and chosen $\{\mathbf{a}_1, ..., \mathbf{a}_m\} \in \mathcal{R}_q^d$ from the uniform distribution, the problem is to find the elements of non-zero integer vector $\mathbf{y} = (y_1, ..., y_m)^T \in \mathcal{R}^m$ such that $(\Sigma_{i=1}^m \mathbf{a}_i \cdot y_i = 0 \mod q)$ and $0 < ||y|| \leq \beta$.

Secret sharing is a technique used in *Threshold cryptography*, in which a threshold scheme involves a dealer who holds a secret, a group of n parties known as shareholders, and a collection

of subsets of parties called the *Access structure*. The secret is divided among the shareholders using a deterministic method that ensures each share reveals no information about the original secret. However, any set in the access structure can reconstruct the secret. Secret sharing is used in various applications of cryptography and distributed computing, such as Byzantine agreement [250], Digital signatures [106], Secure MPC [89], Attribute-based encryption [281], and so on. Shamir [263] and Blakely [38] introduced the first secret sharing schemes. Shamir's scheme is more practical today and is based on *Lagrange polynomial interpolation* [282] while Blakely's scheme is based on *Solving linear systems*.

*Definition 2.1.10 (**Shamir's SS (SSS)**).* Given a secret value s, a polynomial of degree t-1 with random coefficients in $\mathbb{F}_p$, $f(x) = x_0 + x_1x^1 + x_2x^2 + ... + x_{t-1}x^{t-1}$ (x_0=s), is created and n samples are derived, $\{(i_j, f(i_j) \mid j = 1, ..., n\ ,\ i_j \xleftarrow{\$} \mathbb{F}_p\}$, and are given to the parties. To recover the secret, shares of t or more parties are used by Lagrange polynomial interpolation procedure to reconstruct the polynomial $f(x)$ and extract the secret x_0. SSS offers information-theoretic security, ensuring that an adversary with fewer than t shares learns nothing about the secret, as reconstruction requires at least t shares and the missing ones introduce irreducible randomness, regardless of the computational resources. In contrast, computational security assumes computationally bounded adversaries; if this is violated, the scheme is broken.

2.2 Digital Signatures

Digital signatures [79] (RSA [173], ECDSA [9], EdDSA [34]) are essential cryptographic mechanisms that enable scalable authentication with non-repudiation and public verifiability through the use of *PKI*. In digital signatures, a signer S has a key pair (sk, PK) and signs a message using the private key. Anyone possessing the signer's public key can verify the integrity of the message and ensure that it has remained unaltered during transmission.

In addition to standard digital signatures, various specialized (exotic) signatures have been developed to meet different security and functionality requirements [68]. This survey briefly overviews such schemes, focusing only on *group signatures* [76], *ring signatures* [254], and *multi-signatures* [170]. Group signatures enable members of a group to sign messages on behalf of the group while preserving signer anonymity, offering properties such as unforgeability and traceability, useful in organizational settings (e.g., DocuSign's Signing Groups). Ring signatures, a more flexible alternative, do not require predefined groups and allow a signer to anonymously generate a signature on behalf of an arbitrary set (ring) of users. Multi-signature schemes allow multiple parties to jointly generate a single compact signature, ensuring that all signers participate in the signing process. While multi-signatures closely resemble full threshold signatures, group and ring signatures can also be adapted to support threshold variants. We analyze these schemes and their threshold constructions in Sections 8.1–8.3.

Definition 2.2.1 (Digital signature). A digital signature scheme, $SGN = (Setup, Kg, Sig, Ver)$, consists of four algorithms defined as follows:

- $params \leftarrow SGN.Setup()$: Given the security parameter κ, it returns the domain parameters of the scheme.
- $(sk, PK) \leftarrow SGN.Kg(params)$: Given the domain parameters $params$, it outputs the private/public key pair.
- $\sigma \leftarrow SGN.Sig(m, sk)$: Given the message m and the private key sk, it outputs the signature σ.
- $b \leftarrow SGN.Ver(m, , PK)$: Given a message-signature pair (m, σ), and the public key PK, it outputs decision bit $b \in \{0, 1\}$ (if $b = 1$, the signature is valid; otherwise it is invalid).

The domain parameters, *params*, are crucial in determining the intractability assumption and the performance of digital signatures. Further information regarding the domain parameters for standard digital signatures can be found in [79].

2.2.1 Security Models of Digital Signatures. The security model used for digital signatures is *Existential Unforgeability under Chosen Message Attacks (EU-CMA).*

Definition 2.2.2 (EU-CMA). Given a digital signature scheme $SGN = (Setup, Kg, Sig, Ver)$, an attacker (forger) $\mathcal{A}$ with access to a signing oracle $O(sk, .)$, the experiment $Expt_{SGN}^{EU\text{-}CMA}(\mathcal{A})$ is:
$params \leftarrow SGN.Setup()$,
$(sk, PK) \leftarrow SGN.Kg(params)$,
$(m^*, \sigma^*) \leftarrow \mathcal{A}^{O(sk,.), SGN.Sig_{sk}(\vec{m})}(PK)$,
If $SGN.Ver(m^*, \sigma^*, PK) = 1$ & $m^* \notin \vec{m}$, *return* 1 *else* 0.

The *EU-CMA* of $\mathcal{A}$ is defined as: $Adv_{SGN}^{EU\text{-}CMA}(\mathcal{A}) = Pr[Expt_{SGN}^{EU\text{-}CMA}(\mathcal{A}) = 1]$.

The *EU-CMA* advantage of *SGN* is defined as: $Adv_{SGN}^{EU\text{-}CMA}(T, T') = \max_{\mathcal{A}}\{Adv_{SGN}^{EU\text{-}CMA}(\mathcal{A})\}$.

where the maximum is over $\mathcal{A}$ with $\vec{m}$ being the set of queries sent by the $\mathcal{A}$, T the number of queries sent to *SGN* and T' the maximum number of queries sent to $O(sk, .)$.

2.2.2 System Models of Digital Signatures. Digital signatures can be deployed in diverse system models, depending on application-specific requirements and constraints. For example, in IoT environments, signers are often resource-limited devices, whereas verifiers are typically resource-rich. Conversely, in domains like healthcare, servers may act as signers, with verifiers being less powerful desktop computers. In this survey, we do not assume a fixed system model; instead, we focus on the threshold constructions of digital signatures, which can offer varying tradeoffs in security, efficiency, and practicality. The needs and limitations of an application guide the selection of an appropriate threshold scheme.

2.3 Threshold Digital Signatures

Definition 2.3.1 (Threshold digital signature). A (t, n) threshold digital signature, $SGN = (Setup, Kg, Sig, Ver)$, consists of four algorithms defined as follows:

- $\underline{params \leftarrow SGN.Setup()}$: Given the security parameter κ, it returns the domain parameters of the scheme.
- $\underline{(\langle sk_1, sk_2, ..., sk_n\rangle, PK) \leftarrow SGN.Kg(params)}$: Given the domain parameter *params*, it outputs the public key and a set of secret keys, one for each party.
- $\underline{\sigma \leftarrow SGN.Sig(m, \{sk_i\})}$: Given the message m and keys ($|\{sk_i\}| \geq t$), it outputs the signature σ.
- $\underline{b \leftarrow SGN.Ver(m, , PK)}$: Given the message-signature pair (m, σ), and the public key PK, it outputs decision bit $b \in \{0, 1\}$ (if $b = 1$, the signature is valid, otherwise it is invalid).

2.3.1 Security Models of Threshold Digital Signatures. Two security definitions can be used for threshold digital signatures. The *t-private* notion is general to any thresholded scheme, and the *t-EU-CMA* is the thresholded definition of *EU-CMA* mentioned in Definition 2.2.2.

Definition 2.3.2 (t-private protocol). A protocol is *t-private* if, following its execution, any subset of n parties with the size ($< t$) cannot gain more computational knowledge than what they could have obtained individually from their own private data [29].

Definition 2.3.3 (t-EU-CMA). Given a (t, n) digital signature scheme $SGN = (Setup, Kg, Sig, Ver)$, an attacker (forger) $\mathcal{A}$ with access to a signing oracle $O(sk, .)$, the experiment $Expt_{SGN}^{t\text{-}EU\text{-}CMA}(\mathcal{A})$ is as follows:

$params \leftarrow SGN.Setup()$,
$(sk, PK) \leftarrow SGN.Kg(params)$,
$(m^*, \sigma^*) \leftarrow \mathcal{A}^{O(\langle sk_1, sk_2, \ldots, sk_r \rangle, .), SGN.Sig_{\langle sk_1, sk_2, \ldots sk_r \rangle}(\vec{m})}(PK)$ where $r \geq t$,
If $SGN.Ver(m^*, \sigma^*, PK) = 1$ & $m^* \notin \vec{m}$, *return* 1 *else* 0. The *EU-CMA* definitions are as in the Definition 2.2.2.

2.3.2 System Model of Threshold Digital Signatures. The system model of threshold digital signatures generally mirrors that of the underlying single-signer scheme in terms of application use cases, with key distinctions in key generation and signing. While in single-signer settings keys are typically generated by the signer or via an offline trusted setup (e.g., in IoT deployments), threshold schemes use a dealer to distribute the secret key among n parties. Signing requires collaboration among at least t parties, as specified by a (t, n) access structure. The verification process often remains identical (functional interchangeability), using the public key and the original scheme's verification algorithm.

2.4 MPC

(Secure) MPC allows a group of distributed computing parties to jointly compute a function using their private inputs, ensuring that no additional information is disclosed during the computation process except for the function's output. The concept of MPC was initially introduced by Andrew Yao [291] as the "The Millionaires" problem for two parties and later extended to multiple parties by Goldreich et al. [151]. Later, in his subsequent work [292], he presented the *Garbled circuits* protocol as a two-party MPC protocol with a fixed number of rounds involving one party garbling the target function, initially represented as a circuit of logical gates, and sending it to the other party for final computation. Throughout this process, additional communications are needed to assist the other party in computing the function, which is facilitated by techniques like *Oblivious transfer* [161]. Following Yao's protocol, many other protocols in the literature, such as GMW [151], BGW [29], BMR [25], and GESS [186] have been proposed.

Initially, research on MPC primarily concentrated on theoretical aspects, aiming to define and prove the security of presented protocols under different adversarial and network models using complex operations that resulted in excessive overhead. However, numerous early works have contributed to improving the efficiency of MPC constructions via computation algorithm optimizations, memory and communication overhead improvements, and leveraging CPU-level instructions like AES-NI. Subsequent attempts optimized the circuit computation cost. Unlike AND/OR gates, XOR gates are computationally inexpensive [187], and using them in the circuit can significantly improve the performance. While constructing the circuits manually [169] was challenging, researchers decided to use *Special-purpose MPC Compilers* [162] to generate efficient circuits for the functions written in the ***Secure Function Definition Language (SFDL)*** language.

The security evaluation of MPC protocols involves considering factors such as the adversary's behavior, corruption strategy, the number of compromised parties, and the execution environment:

Adversary's behavior : A **Semi-honest (passive)** adversary is an attacker who corrupts parties and acquires their internal state, including private input and message transcripts, while adhering to the protocol. In contrast, a **Malicious (active)** adversary may force corrupted parties to deviate from the protocol. An adversary is considered **Covert** if they attempt to breach the protocol. This security breach can be detected and adjusted with a specific probability.

Parties' corruption: **Static** corruption refers to a scenario in which honest parties maintain their honesty throughout the entire protocol execution, and the corrupted parties remain unchanged for the whole duration of the protocol execution. **Adaptive** corruption, on the other hand, occurs when the adversary, based on the knowledge acquired, selects which parties to corrupt. **Proactive**

corruption involves a situation where certain parties can be corrupted while the protocol progresses. However, these corrupted parties remain compromised for a specified duration and return to an honest state once a security breach is detected.

Number of corrupted parties: When the adversary's ability to corrupt is limited to fewer than half of the parties, the security model is characterized as **honest majority**. Conversely, if the adversary can corrupt more than or equal to half of the parties, the model is **dishonest majority**.

2.4.1 ***Universally Composable (UC) Security Framework.*** A central challenge in cryptographic protocol design is ensuring security under concurrent execution, as real-world systems rarely run protocols in isolation. Early models focused on stand-alone settings, offering clean analysis but failing to capture composability, i.e., the preservation of security when protocols interact with each other. The UC security framework [65] addresses this by enabling security analysis in concurrent environments. Security is defined with respect to an *ideal functionality*, and a protocol *UC-realizes* it if, for every real-world adversary, there exists a simulator in the ideal world such that no environment can distinguish between the two executions. A key strength of the UC framework is its *Universal Composition Theorem*, which guarantees that a protocol proven UC-secure can replace the ideal functionality in any larger system without compromising overall security. This composability facilitates modular protocol design, and the simulation-based approach ensures robustness in adversarial, dynamic, and networked environments.

3 Threshold Schnorr-like Digital Signatures and Standards

The NIST [79] has standardized digital signatures that rely on three well-known security problems: IF, DLP, and ECDLP (see Definitions 2.1.1, 2.1.2, and 2.1.3). In the upcoming sections, we will explore thresholding of *Schnorr* signature [56] based on the DLP and its (EC)-based variants (see Figure 2), *EdDSA* [56, 174] and *ECDSA* [79] based on the ECDLP.

3.1 Threshold Schnorr-based Signatures

The Schnorr signature, introduced by Claus Schnorr [262], is a DLP-based digital signature. Various existing standards, such as ECDSA and EdDSA, are EC-based variants of the Schnorr signature. A seminal work of Horster et al. [166] integrated a broad class of approaches regarding Schnorr-like and Elgamal-like signatures and presented a general *Meta-ElGamal* family. Inspired by this Meta-Elgamal family, we present Figure 2 to illustrate the relationship between Schnorr-like signatures. Specifically, the comparison of the operations of the original Schnorr scheme operating on integer groups and EC-Schnorr variants, along with their algorithms, has been illustrated in Figure 2(a) and (b), respectively. Table 3 contains a compilation of recent works on threshold Schnorr signatures.

Benhamouda et al. [32] proposed high-throughput threshold Schnorr signature protocols that enable parallel signing of multiple messages within a single execution. These protocols are designed for both synchronous and asynchronous settings, with a particular emphasis on distributed (t, n) signing over asynchronous public blockchains. A central challenge in this context is the computation of the value R, which they address by employing ***Distributed Key Generation (DKG)*** [145] to securely share the nonce r among participants. Unlike conventional DKG approaches that require complete secret sharing, their protocol allows progress as soon as a sufficient number of honest parties possess the necessary shares, thereby improving efficiency under partial availability. To accelerate the generation of ephemeral secrets required for repeated DKG executions, they utilize *packed secret sharing* [136] and *super-invertible matrices* [165], trading off some resiliency in the threshold parameter t. Furthermore, instead of relying on traditional interactive MPC protocols, they introduce a non-interactive design by pre-sharing long-term secret packs, enabling SIMD-style partial signature generation through local multiplications. This eliminates the need for interaction

(a) (DLP and ECDLP)-based Schnorr Signature Comparison [56]

Phase	Original Schnorr	Elliptic-curve Schnorr
Key-Gen	$PK \leftarrow g^{sk} \bmod p$	$PK \leftarrow sk \times B$
Commit nonce	$R \leftarrow g^{r} \bmod p$	$R \leftarrow r \times B$
Compute challenge	$e \leftarrow H(R\|\|PK\|\|m)$	$e \leftarrow H(R\|\|PK\|\|m)$
Signature generation	$s \leftarrow r - e \cdot sk \bmod q$	$s \leftarrow r + e \cdot sk \bmod n$
Signature verification	$R \stackrel{?}{=} PK^{e} \cdot g^{s} \bmod q$	$s \times B \stackrel{?}{=} R + c \times PK$

(b) Variants of Schnorr Signature Scheme

Schnorr Digital Signature

params $\leftarrow$ *Setup*(κ):
1: Generate p and q s.t. $p > q$ and $p|q-1$, set G as $\mathbb{Z}_p^*$ and g as its generator
2: **return** (p, q, G, g)

$(sk, PK) \leftarrow Kg(params)$:
1: $sk \xleftarrow{\$} \mathbb{Z}_q^*$, $PK \leftarrow g^{sk} \bmod p$
2: **return** (sk, PK)

$\sigma \leftarrow Sig(m)$:
1: $r \xleftarrow{\$} Z_q^*$, $R \leftarrow g^{r} \bmod p$
2: $e \leftarrow H(R||m)$, $s \leftarrow (r - e \cdot sk) \bmod q$
3: **return** (R, s)

$\{0, 1\} \leftarrow Ver(\sigma, m, PK)$:
1: $R' \leftarrow PK^{e} \cdot g^{s} \bmod q$
2: **return** $R' \stackrel{?}{=} R$

ECDSA Digital signature

params $\leftarrow$ *Setup*(κ): **return** $(CURV, B, n)$

$(sk, PK) \leftarrow Kg(params)$:
1: $sk \xleftarrow{\$} Z_n^*$, $PK \leftarrow sk \times B$
2: **return** (sk, PK)

$\sigma \leftarrow Sig(m)$:
1: $k \xleftarrow{\$} \mathbb{Z}_n^*$, $(x_1, y_1) = k \times B$, $r \leftarrow (x_1 \bmod n)$
2: $s \leftarrow k^{-1}(\mathrm{H(m)} + r \cdot sk) \bmod n$, if s is 0 goto 1
3: **return** (r, s)

$\{0, 1\} \leftarrow Ver(\sigma, m, PK)$:
1: $w \leftarrow \sigma^{-1} \bmod n$
2: $u_1 \leftarrow (H(m) \cdot w \bmod n)$, $u_2 \leftarrow (r \cdot w \bmod n)$
3: $(x_1, y_1) \leftarrow u_1 \times B + u_2 \times PK$
4: If (x_1, y_1) is O, **return** 0
5: **return** $r \stackrel{?}{=} (x_1 \bmod n)$

EdDSA Digital signature

params $\leftarrow$ *Setup*(κ): **return** $(CURV, B, n)$

$(sk, PK) \leftarrow Kg(params)$: The private key sk is a b-bit random byte string. The hash function H, produces a digest of size $2b$.
1: $sk \xleftarrow{\$} Z_q^*$, $(h_0, h_1, ..., h_{2b-1}) \leftarrow H(sk)$
2: $a \leftarrow (h_0, ..., h_{b-1})$, $b \leftarrow (h_b, ..., h_{2b-1})$

$\sigma \leftarrow Sig(m)$:
1: $r \leftarrow H(m||b)$, $R \leftarrow r \times B$
2: $s \leftarrow r + \mathrm{H(R||PK||m)} \cdot sk \bmod n$
3: **return** (R, s)

$\{0, 1\} \leftarrow Ver(\sigma, m, PK)$:
1: $h \leftarrow H(R, m, PK)$
2: **return** $8sB \stackrel{?}{=} 8hPK + 8R$

(c) FROST Signature Scheme [188]

FROST Key Generation

Round 1

Party P_i:
1. Samples $(a_{i0}, a_{i1}, ..., a_{i(t-1)}) \xleftarrow{\$} \mathbb{Z}_q$ and constructs polynomial $\Sigma_{j=0}^{t-1} a_{ij} x^j$.
2. Computes proof of knowledge to the secret a_{i0}:
   - $k \xleftarrow{\$} \mathbb{Z}_q$, $R_i \leftarrow g^k$
   - $c_i = H(i, \phi, g^{a_{i0}}, R_i)$, $\mu_i = k + a_{i0} \cdot c_i$ where ϕ is context string.
   - $\sigma_i \leftarrow (R_i, \mu_i)$
3. Computes and sends the public commitment $\overrightarrow{C}_i = \langle \phi_{i0}, ..., \phi_{i(t-1)} \rangle$ where $\phi_{ij} = g^{a_{ij}}$, $0 \le j \le t-1$ along with σ_i
4. Receives $\overrightarrow{C}_l$ and σ_l of other parties and checks:
   - $\sigma_j \stackrel{?}{=} (R_j, \mu_j)$ and $R_j \stackrel{?}{=} g^{\mu_j} . \phi_{j0}^{-c_j}$

Round 2

1. Securely sends each party P_j a secret share $(j, f_i(j))$ and then deletes it except for $j = i$
2. Verifies their shares by $g^{f_j(i)} \stackrel{?}{=} \Pi_{k=0}^{t-1} \phi_{jk}^{i^k \bmod q}$ and aborts if it fails.
3. Calculates long-lived private signing shares $s_i = \Sigma_{j=1}^{n} f_j(i)$
4. Computes their public verification share $Y_i = g^{s_i}$ and the group's public key $Y = \Pi_{j=1}^{n} \phi_{j0}$

FROST Signature Generation

Phase 1

Signature aggregator ($\mathcal{SA}$):
1. Fetches next commitment for each party $P_i \in S$ from L_i and constructs B.
2. Sends each party the tuple (m, B).

Phase 2

Party P_i:
1. Validates the message m and checks $D_j, E_j \in Z^*$ for each commitment in B.
2. Computes binding value set $\rho_j = H_1(j, m, B)$ for each party j, $R = \Pi_j D_{j \in S} . (E_j)^{\rho_j}$, and $c = H_2(R, Y, m)$.
3. Computes their response as $z_i \leftarrow d_i + (e_i . \rho_i) + \lambda_i . s_i . c$ and sends it to the $\mathcal{SA}$.

Phase 3

Signature aggregator ($\mathcal{SA}$):
1. Derives $\rho_i = H_1(i, m, B)$, $R_i = D_{ij} . (E_{ij})^{\rho_i}$ for each party i and further $R = \Pi_{i \in S} R_i$ and $c = H_2(R, Y, m)$.
2. Verifies $g^{z_i} \stackrel{?}{=} R_i . Y_i^{c . \lambda_i}$.
3. Computes group's response $z \leftarrow \Sigma z_i$.
4. Returns $\sigma \leftarrow (R, z)$.

Fig. 2. Schnorr signature and its variants.

Table 3. Summary of Threshold Schnorr Digital Signatures

Work by	Description
BDLR25 [17]	➥ Constructing Glacius, the first adaptively secure threshold Schnorr signature with full corruption threshold, constant key size, and identifiable abort ➥ Achieves adaptive security under the standard DDH assumption in the Random Oracle Model ➥ Supports full corruption threshold $t < n$ in contrast to prior schemes which were limited to $t \leq \frac{n}{2}$ ⋆ Uses constant-size keys (three field elements) and requires only five signing rounds, while maintaining security and efficiency
KG25 [189]	➥ Introducing Arctic, the first efficient, deterministic, and stateless two-round threshold Schnorr signature scheme ➥ Eliminates secret state across rounds, mitigating key-recovery attacks from nonce reuse ➥ Introduces **Verifiable Pseudorandom Secret Sharing (VPSS1)** for public nonce generation without zero-knowledge proofs ➥ Maintains compatibility with standard Schnorr verification ⋆ Only two communication rounds ⋆ Over 10× speedup for $n \leq 20$ and up to 1,000× for $n \leq 10$ vs. prior deterministic schemes ⋆ Highly parallelizable; scalable for moderately sized groups
KOR23 [190]	➥ Proposes a stateless threshold Schnorr signing approach using **Pseudorandom Correlation Functions (PCFs)** [54], enabling distributed derandomization ➥ Presents two instantiations based on ***Vector Oblivious Linear Evaluation (VOLE)*** correlations: (i) A two-round protocol from [256] achieving covert security under PRF and OT assumptions; (ii) A two-round protocol based on Paillier encryption-based PCF [240] achieving full malicious security under **Decisional Composite Residuosity (DCR)** and Strong RSA assumptions ⋆ Bandwidth overhead: 0.1 KB (first scheme), 0.5 KB (second scheme)
JPS22 [175]	➥ Proposing ATSSIA, a non-interactive asynchronous threshold Schnorr signature ➥ Improving the scheme [188] by having the signing operation running independent of the parties and not requiring the signers to be online all in once until the aggregation phase ➥ Compared to [188], it is robust, does not have a preprocessing stage and allows dynamic participation ➥ Secure against ROS solver [33] and Drijvers et al. [112] attacks in ROM
RRJSS22 [257]	➥ Proposing ROAST, a wrapper that takes in a semi-interactive threshold signature scheme (i.e., signing consists of preprocessing and signing rounds) and converts it to a protocol that is robust and asynchronous ➥ Applying the wrapper to the FROST scheme and obtaining an efficient and robust Schnorr signature scheme ➥ Providing robustness through having concurrent signing sessions and reduce the exponential number of the sessions from $\binom{n}{t}$ to at most $n - t + 1$ ⋆ Using FROST for benchmark with (67, 100) threshold setting running in 3 seconds⋆ Possible improvements by supporting a group of multiple input messages simultaneously, allowing continuous stream signing, and letting signers be in more than one session to reduce latency
BGLM22 [22]	➥ Proposing a (2, 3) threshold Schnorr-based variant of [24] with an offline recovery participant for crypto-assets custody ➥ Provable security against adaptive adversaries ➥ Capable of achieving resiliency of recovery in the presence of a malicious adversary if the encryption scheme supports DLP verification using ZKP ➥ Using an IND-CPA [217] encryption scheme for communicating with the offline party along with a secure non-malleable commitment scheme ➥ Using **ZKP of Knowledge (ZKPoK)** to prove the binding of the secret values and their corresponding public values ➥ Using Feldman's VSS [128] for the secret sharing

➥ Technical aspects of the scheme ⋆ Additional notes such as performance analysis.

during signing and significantly improves throughput. Additional optimizations include a modular QUAL agreement protocol over broadcast channels and a randomness beacon for sub-sampling committee members. As a result, their scheme achieves $\Omega(n^2)$ signatures per execution with $O(n^2)$ broadcast bandwidth, where n denotes the committee size.

Komlo et al. [188] introduced FROST, a round-optimized threshold Schnorr signature scheme, with its signature operations illustrated in Figure 2(c). Unlike traditional threshold signatures that emphasize robustness by identifying and excluding misbehaving parties, FROST adopts a more optimistic and efficiency-focused approach: if misbehavior is detected, the protocol simply aborts and restarts, avoiding the overhead of fault resolution. To reduce communication, FROST incorporates a semi-trusted signature aggregator ($\mathcal{SA}$), which may be one of the parties or an external entity. Although $\mathcal{SA}$ facilitates aggregation, it neither exposes the private key nor enables forgery if compromised. In the key generation phase, FROST employs a variant of Pedersen's DKG [144] based on Feldman's VSS [128], where each party's share is the sum of those received from all VSS instances. To mitigate rogue-key attacks [26] when $t \geq n/2$, each party must prove knowledge of their secret share in zero-knowledge. The signing process consists of two integrated phases. The preprocessing phase enhances security against Drijvers et al.'s attacks [112] by binding each party's response to a specific message and subset of committed participants, using non-interactively generated distributed nonces. This ensures both efficiency and robustness. The subsequent signing phase, also shown in Figure 2(c)(right), relies on additive secret sharing and merges seamlessly with the preprocessing phase, yielding a unified two-round protocol.

Building on prior works, threshold Schnorr signatures continue to evolve with significant recent developments [19]. Crites et al. [92] demonstrate a plausible attack, indicating that schemes like FROST and Sparkle [91] may not achieve full adaptive security under standard assumptions. To address such limitations, Bacho et al. [16] introduce HARTS, a threshold Schnorr that achieves adaptive security, asynchronous robustness, and arbitrary threshold support, tolerating up to $t < n/3$ corruptions and completing signing in a single asynchronous round with near-optimal communication. Mergoupis-Anagnou et al. [221] propose OSST, a non-interactive Schnorr-based threshold identification protocol enabling any t or more parties to prove knowledge of a secret

Table 4. Summary of Threshold ECDSA Digital Signatures

Work by	Description
LLZXW+25 [210]	➺ Proposing two three-round robust threshold ECDSA, RompSig-Q and RompSig-L, which have two offline broadcasts and one non-interactive online phase, achieving round efficiency with robustness and identifiable abort ➺ RompSig-Q uses non-interactive DKG based on Kate et al. [180], vectorized **Multiplicative-to-Additive (MtA)** via **Castagnos-Laguillaumie (CL)** encryption, and Fiat–Shamir-based NIZKs to achieve signing with bandwidth-efficient broadcasts ➺ RompSig-L supports dynamic participation ⋆ RompSig-Q emphasizes online latency and RompSig-L targets scalability, achieving best tradeoffs in threshold ECDSA
THLWY24 [271]	➺ Introducing a new Multi-Dimension Forking Lemma that enables efficient batch range proofs for Paillier and JL encryptions, addressing a key bottleneck in HE-based threshold ECDSA protocols ➺ Constructing efficient batch MtA protocols using the new lemma, improving amortized bandwidth and computation by 2–3× over prior works ➺ Demonstrating concrete improvements across several existing threshold ECDSA schemes, including 2.1–2.4× bandwidth and 1.5–1.7× computation savings over [66, 110, 286] ⋆ Enabling more practical deployment of HE-based threshold ECDSA in settings like blockchain wallets and distributed key management by reducing cost of range proofs, which account for over 75% of presigning overhead.
DKLS24 [110]	➺ Proposing a three-round threshold ECDSA protocol that achieves malicious security against a dishonest majority, while UC-realizing the standard threshold signing functionality ➺ Constructing the protocol using only ideal commitment and two-party multiplication (VOLE) primitives, with no zero-knowledge proofs and a simple commit-release-and-complain mechanism for key generation ➺ Introducing a novel, efficient fused double-instance of Bar-Ilan and Beaver's inverse computation protocol [20], with statistical consistency checks that reuse correlated randomness as implicit MACs ⋆ Achieving best-in-class efficiency, outperforming prior OT- and Paillier-based schemes by 6× in high-latency environments, while matching the round complexity of threshold Schnorr
XALCC+23 [286]	➺ Proposes a new MtA functionality based on the **Joye–Libert (JL)** cryptosystem [177], improving both computation and communication in threshold ECDSA signing ➺ Constructs a modified JL cryptosystem over $\mathbb{Z}_n$ (instead of $\mathbb{Z}_{n^2}$ as in Paillier), based on k-quadratic residuosity and Strong RSA assumptions ➺ Introduces a JL-based commitment scheme supporting compact scalar/vector commitments and efficient public operations (e.g., ciphertext exponentiation) ➺ Designs the first standard-sound zero-knowledge proofs for JL encryption, including range, equality, and affine operation proofs ⋆ Achieves 1.85–2× lower bandwidth and 1.2–1.7× faster computation than Paillier-based; 7× faster than CL-based at 2× bandwidth cost; 18.7× bandwidth savings vs. OT-based methods
BLMS22 [24]	➺ Proposing the first $(2, 3)$ threshold ECDSA scheme with allowing the presence of one offline party for signature recovery ➺ With four possible phases: *Setup* phase played by all parties exchanging public parameters, *Key generation* played by two players only with third one already chose a long-term asymmetric key and went offline, *Signing* Played by two players, and *Signature recovery* Played with the third player and one of the other ones ➺ Proven security against adaptive security based on standard assumptions ➺ Requiring provable encryption scheme instead of a CPA asymmetric encryption to deal with malicious offline party
DJNPO22 [95]	➺ Proposing a bandwidth efficient four-round (t, n) threshold ECDSA scheme with two phases of preprocessing (three rounds) and signing phase (one round) ➺ Proving secure against active adversary in the honest majority model for $n \geq 2t + 1$ and with a UC security proof ➺ Using Shamir's secret sharing over Pedersen's verifiable secret sharing for joint random secret sharing of the private key ➺ Extending the proposed method to support fairness and termination ⋆ Achieving $n \log q$ bits communication with two long curve multiplication in signing, $6n \log q + 2n \log p$ bits communication with three long curve multiplication in presigning and $n \log q + n \log p$ bits communication in the key generation phase

➺ Technical aspects of the scheme ⋆ Additional notes such as performance analysis.

without revealing it or requiring public shares. Separately, Bacho et al. [17] construct a three-round adaptively secure scheme from DDH, while Cimatti et al. [85] develop a dynamic threshold Schnorr supporting updates to both the signer set and threshold without a trusted party. Meanwhile, Chen [80] presents two round-efficient schemes with rewinding-based adaptive security proofs. Finally, Shoup [267] introduces a framework that abstracts and unifies core techniques for constructing threshold Schnorr signatures.

3.2 Threshold Elliptic Curve Digital Signature Algorithm (ECDSA)

The ECDSA, shown in Figure 2(b), is a Schnorr-based variant that employs ECs and is standardized by ANSI [9] and NIST [79]. A notable variant by Pornin [248] introduces deterministic signing. ECDSA is widely deployed in security-critical applications such as DNSSEC [93], TLS, and cryptocurrencies like Bitcoin [150]. Owing to its broad adoption, considerable research has focused on constructing two-party and multi-party threshold ECDSA, aiming to reduce communication rounds, lower overhead, and improve signing latency, while introducing a trustless setup and identifiable aborts. Tables 4 and 5 summarize recent foundational works.

To construct a threshold ECDSA scheme [108, 140, 142, 201, 204], two key components must be distributed: the private key sk and the nonce k. However, securely computing the nonce commitment R and the product $k \cdot sk$, required to derive the signature component s, presents significant challenges. Early schemes, such as those of Gennaro et al. [142], addressed this issue using Paillier's cryptosystem, but suffered from high overhead due to the long ciphertexts and costly zero-knowledge proofs. To improve efficiency, Gennaro et al. [140] adopted the SPDZ framework [96] and employed a two-party protocol by Gilboa [148], which transforms multiplicative shares into additive ones.

Castagnos et al. [69] proposed improvements to the threshold ECDSA scheme by Gennaro et al. [142], addressing limitations of Paillier's cryptosystem, particularly under malicious adversaries. Specifically, the mismatch between Paillier plaintexts in $(\mathbb{Z}/N\mathbb{Z})$ and ECDSA signatures in $(\mathbb{Z}/q\mathbb{Z})$ necessitated a large N to prevent wraparound, increasing computational cost and

Table 5. Summary of Threshold ECDSA Digital Signatures (Cont.)

Work by	Description
XAXYC21 [287]	➥ Proposing a new online-friendly two-party threshold ECDSA protocol that reduces the complxity of offline computation while keeping the online phase optimal and non-interactive ➥ Uses only one call to the MtA functionality requiring heavy and complex computation compared to previous works needing 2-4 calls ➥ Proposing linear nonce sharing for improving flexibility compared to additive/multiplicative sharing and signing private key resharing for improving alignment with the optimized MtA-based computation ➥ Generalizable to 2-out-of-n threshold ECDSA using SSS, maintaining the same online complexity ⋆ Non-interactive signing with one call to MtA ⋆ Implementation in Rust with MtA instantiation with Paillier encryption, CL encryption, and oblivious transfer
CGGMP20 [66]	➥ Proposing two non-interactive threshold ECDSA schemes with the signing procedure having a (3-6)-round preprocessing phased before receiving the message and a non-interactive signing phase in which parties generate their signature shares ➥ Providing proactive security for the key refreshment ➥ Providing the capability of identifying the party deviating from the protocol with aborting the signature generation ➥ Achieving UC security in ROM ⋆ Small number of rounds compared to [109, 140, 204] ⋆ Higher communication compared to [140, 204] with Paillier and less compared to [109, 204] with oblivious transfer
DKLS19 [109]	➥ Proposing a (t, n) threshold ECDSA scheme with $\log t + 6$ rounds of computations ➥ Proven security against malicious adversary with the ***Computational Diffie-Hellman (CDH)*** *Assumption* ➥ Improving the setup model of [108] with security against dishonest majority ➥ Proposing a two-party multiplication protocol based on oblivious transfer, improving the one in [108] by 40% through requiring parties to choose random inputs and later correcting them if necessary ➥ Presenting a technique to extend two-party multiplier to a t-party multiplier with $\log t$+(1 or 2) rounds ➥ Constructing a signing procedure consisting of three t-party multipliers responsible for converting shares from multiplicative into an additive and producing an additive sharing from product of two additive shares ⋆ Giving benchmarks for LAN, WAN, and resource-limited devices ⋆ Achieving significant better performance for setup phase compared to [44, 142], weaker performance for $(2, n)$ signing phase compared to [108] and same performance for $(2, n = 24)$ with parallelism compared to $(2, 2)$ of [201]

➥ Technical aspects of the scheme ⋆ Additional notes such as performance analysis.

requiring range proofs to verify encrypted values. To mitigate this, they leveraged their earlier technique [70], which uses hash-proof systems and Castagnos and Laguillaumie's homomorphic encryption scheme [72], instantiated over class groups of imaginary quadratic fields. This allowed the message space to align with $(\mathbb{Z}/q\mathbb{Z})$, eliminating the need for large moduli and range proofs. Additional refinements ensured ciphertext well-formedness and addressed non-surjectivity in the encryption scheme. In follow-up work, Castagnos et al. [71] extended this approach to construct a threshold ECDSA scheme with accountability, online/offline efficiency, and adaptive security in the (n, n) setting. Building on the online/offline paradigm of Gennaro et al. [141] (later formalized in [66]), they shifted expensive operations, specifically the joint computation of $k^{-1} \cdot sk$ and k, to the offline phase. In the online phase, each party computes a signature share $s_i = k_i^{-1}(H(m) + r \cdot sk_i)$ non-interactively, which is then aggregated to form the final signature s. While zero-knowledge proofs can enforce accountability, their use introduces complexity. Instead, the approach in [141] detects malicious behavior by checking public values s_i and randomness revealed during the offline phase. However, this cannot be directly applied to [70] due to the use of hash-proof systems. To resolve this, new zero-knowledge proofs were introduced that allow simulation without compromising efficiency.

Threshold ECDSA, akin to threshold Schnorr, has remained an active area of research. Recent work by Koziel et al. [191] presents a proactive two-party scheme improving upon [201] by enabling faster key generation and efficient secret-share refreshment without regenerating the public key. Their design replaces costly Paillier ZK proofs with a lightweight weak proof, enhancing practical efficiency while supporting proactive security. Complementarily, Tang et al. [273] propose the first three-round threshold ECDSA scheme that is robust and online-friendly, achieving fast signing using only EC operations, yielding an online phase 2.5× faster than [284] and orders of magnitude faster than [283].

3.3 Threshold EdDSA

The EdDSA digital signature, depicted in Figure 2(b), proposed by Bernstein et al. [34] and specified in [174], is a digital signature that utilizes Edwards curves and is based on the ECDLP (See Definition 2.1.3). It is a variant of the Schnorr signature and utilizes performance-optimized ECs specified by the NIST [56], which include the Edwards25519 curve for 128-bit strength and the Edwards448 curve for 224-bit strength. These ECs support two distinct signing modes, each determined by whether the message undergoes prehashing or not, resulting in a total of four modes. Table 6 contains a compilation of recent works on threshold EdDSA signatures.

Battagliola et al. [23] proposed a multi-party EdDSA signature scheme that is provably secure against adaptive adversaries, requiring no trusted party and allowing partial participation during

Table 6. Summary of Threshold EdDSA Digital Signatures

Work by	Description
FYZWY+25 [132]	➺ Proposing a new stateless, deterministic multi-party EdDSA protocol in the full-threshold setting, tolerating all-but-one malicious corruption and reducing communication by 56× compared to [139] with a 2.25× increase in computational cost, while maintaining a 3-round design ➺ Introducing a **multi-verifier zero-knowledge** (**MVZK**) protocol for nonce generation using PCFs and refined multi-verifier extended doubly-authenticated bits (mv-edaBits) ⋆ Ensuring nonce correctness with efficient IT-MAC-based ZK proofs, avoiding MPC overheads, and enabling practical, secure EdDSA signing for blockchain wallets and custody systems.
XFZW+25 [285]	➺ Proposing TAPS-PR, an EdDSA-based threshold signature protocol achieving accountability, privacy, and proactive key refresh, where the threshold is variable and hidden from the public, and misbehavior can be traced via zero-knowledge proofs ➺ Supporting secure key updates that maintain public keys unchanged, ensuring epoch consistency and long-term key protection ➺ Introducing ATS-PR, a more efficient variant of TAPS-PR with reduced communication and computation costs, suitable for financial scenarios prioritizing efficiency over privacy ⋆ TAPS-PR and ATS-PR improve over prior works in computation and bandwidth, with ATS-PR offering the best performance tradeoff in EdDSA-based blockchain systems like Tezos.
FYMH23 [131]	➺ Proposing two practical multi-party EdDSA signature schemes for semi-honest and malicious security where the semi-honest protocol realizes the EdDSA standard and the malicious protocol provides security against a static adversary with at most $n-1$ corrupted parties and eliminates the distributed hashing and the MPC required for it by securely managing a global state ➺ Extending the malicious protocol to allow identifiable abort ⋆ Implementing the proposed protocols in C++ based on Ed25519 curve and running them with 2-5 parties on Alibaba cloud servers ⋆ 14.47 and 4.13 milliseconds for key generation and 15.3 and 10.26 milliseconds for signing in semi-honest and malicious settings with five parties, respectively.
SKKN+22 [264]	➺ Proposing a (t, n) EdDSA threshold signature scheme proved to be secure based on the ECDLP problem and consisting of two phases: Setup phase, in which the private key and secret vectors are secretly shared among the parties. Signing phase, in which using distributed random element generation approaches, parties can generate signatures efficiently without using ZKP and MPC ⋆ Providing performance comparison both for the cloud and embedded implementations for the number of parties and threshold values up to 9.
BLMS20 [23]	Proposing a $(2, 3)$ EdDSA digital signature scheme with an offline recovery party provably secure against an adaptive adversary using a non-malleable commitment scheme, a secure hash function, and an IND-CPA encryption scheme (See Section 3.3 for details).
FHLM+20 [130]	➺ Proposing the first two-party EdDSA threshold signature scheme with a security reduction to the security of the original EdDSA scheme ⋆ Providing performance analysis for signature and key generations in client/server and device/device models for semi-honest and malicious security settings.

➺ Technical aspects of the scheme ⋆ Additional notes such as performance analysis.

key generation. The construction extends their earlier threshold ECDSA scheme [24], which achieves resilience against rushing and malicious adversaries. The proposed $(2, 3)$ threshold scheme employs a non-malleable commitment scheme, a secure hash function, and an IND-CPA encryption scheme. After an initial setup and exchange of common parameters, one party selects a long-term asymmetric key and may go offline. Key generation can proceed between the remaining two parties or between the offline party and one of them. Signing requires the collaboration of two parties, with determinism ensured via Purify [232], an EC-based PRF used to derive verifiable nonce values. If one party becomes unavailable during signing, the offline party can rejoin to enable signature recovery.

4 Threshold Pairing-based Digital Signatures

In this section, we will briefly explore the concept of pairing-based cryptography and introduce a pairing-based signature known as BLS [48]. We will then describe how this signature scheme can be thresholded.

Definition 4.0.1 (Pairings). Let (G_1, g_1) and (G_2, g_2) represent two additive groups with their generators, and G_T be a multiplicative group of prime order q. We introduce $\hat{e}$ as a *bilinear map* or *pairing*, denoted as $\hat{e} : G_1 \times G_2 \to G_T$, possessing the following properties:

— *Bilinearity*: $\forall g_1 \in G_1, \forall g_2 \in G_2$, and $\forall a, b \in \mathbb{Z}_q^* \to \hat{e}(ag_1, bg_2) = \hat{e}(g_1, g_2)^{ab}$ holds.
— *Non-degeneracy*: $\forall g_1 \in G_1, \forall g_2 \in G_2$, and $g_1 \neq 0$ and $g_2 \neq 0 \to \hat{e}(g_1, g_2)$ generates G_T.
— *Computability*: Computation of $\hat{e}$ should be efficient.

Notable pairings in this context are the *Weil pairing* [220] and the *Tate pairing* [137]. As pairings advanced, they found applications in various cryptographic domains. For instance, pairings were utilized to create Identity-based Encryption [43] and PKE with Keyword Search [41]. In digital signatures, the **Boneh, Lynn, and Shacham** (**BLS**) signature scheme [48] was introduced and functions in any group with an easily solvable DDH and a computationally hard CDH. Additionally, pairings played a role in key agreement protocols, including the one-round 3-party key agreement scheme by Joux [176] and its multi-party extension by Barua et al. [21].

Definition 4.0.2 (BLS Digital Signature). Given a computable non-degenerate pairing $\hat{e} = G_1 \times G_2 \rightarrow G_T$ in groups G_1, G_2, and G_T of prime order q with g_1, g_2, and g_t the generators of them, respectively. With $H : \mathcal{M} \rightarrow G_1$ be a Map-to-point hash function, we define the $BLS = (Setup, Kg, Sig, Ver)$ signature as:

- $\underline{params \leftarrow BLS.Setup(\kappa)}$: Setting up the groups G_1, G_2, and G_T.
- $\underline{(sk, PK) \leftarrow BLS.Kg(params)}$: Choose $sk \xleftarrow{\$} \mathbb{Z}_q$ and compute $PK \leftarrow g_2^{sk} \in G_2$.
- $\underline{\sigma \leftarrow BLS.Sig(m)}$: Given the message m, compute $h \leftarrow H(m) \in G_1$ and the signature $\sigma \leftarrow h^{sk}$.
- $\underline{b \leftarrow BLS.Ver(m, \sigma, PK)}$: Given the message-signature pair (m, σ) and the public key PK, check if $\hat{e}(\sigma, g_2) \stackrel{?}{=} \hat{e}(H(m), PK)$ holds and output the decision bit b ($b = 1$ if the statements is true, otherwise 0).

A thresholded version of the BLS digital signature was first given by Boldyreva [39]. She proposed a (t, n) threshold digital signature that can tolerate any $t < n/2$ malicious parties proven in ROM with the capability of achieving proactive security using [164]. Moreover, the key generation process does not require a trusted dealer, and the signature generation is non-interactive and does not utilize any ZKP. Given the groups G_1, G_2, and G_T, the key generation is done using the DKG [145], which is jointly executed by a set of parties. This results in outputting a public key $y = g^x$, and giving each party a private key share x_i such that x can be constructed using Shamir's secret sharing as $x \leftarrow \Sigma_{i \in R}(L_i x_i)$ where L_i is Lagrange coefficient and R is a set of $t + 1$ players. To generate a signature, each party computes and broadcasts a signature share $\sigma_i \leftarrow H(m)^{x_i}$, independent of the number of parties. The signature can be constructed by any player(s) having shares of R as $\sigma \leftarrow \Pi_{i \in R}(\sigma_i^{L_i})$. The verification remains the same as the BLS signature.

Several works have advanced threshold adaptations of the BLS signature scheme, addressing various aspects such as adaptive security and efficiency. Early efforts by Bacho et al.[18] tackled the lack of adaptive security proofs in Boldyreva's original construction [39], while Tomescu et al.[274] proposed a (t, n) threshold BLS scheme with $\mathcal{O}(t \log^2 t)$ aggregation time, maintaining constant signature size, signing, and verification time. Bellare et al.[28] introduced a framework for non-interactive threshold signatures, formalizing **unforgeability** (**UF**) and **strong unforgeability** (**SUF**), and applied it to both threshold BLS and FROST. Garg et al. [138] contributed a threshold BLS construction with proactive and forward security, supported by a silent setup.

Building on these foundations, recent efforts have advanced adaptively secure threshold BLS signature schemes. Das et al. [100] introduced the first construction provably secure under standard assumptions in the ROM model. Their scheme supports non-interactive signing, signature uniqueness, functional interchangeability, and compact output, with only modest overhead (where each secret key share expands to three field elements versus one in [39]), while verification and communication remain efficient. Building on this, Libert [200] proposed a variant that retains core properties while reducing secret key share size by 33%, partial proof cost by 25%, and public key size. Additional works [134, 171] further explore adaptive security for threshold BLS and assess compatibility with BUFF security [90].

5 Threshold RSA Digital Signature

The RSA digital signature [79, 226], based on the IF problem (see Section 2.1), has several known threshold constructions [8, 94] (see Table 7). However, compared to EC-based schemes, threshold RSA has seen limited practical use in recent years.

Table 7. Summary of Threshold RSA Digital Signatures

Work by	Description
CLKH+23 [74]	➺ Proposing a threshold RSA-based aggregate signature for blockchain that minimizes block size and network overhead by replacing all transaction signatures in a block with a single constant-sized aggregate signature ➺ Introducing an accountable subgroup mechanism, supporting low-interaction signing, and reducing the number of exchanged messages to $t + n$ regardless of network size ⋆ Reduces verification cost and storage load, suitable for high throughput applications.
DLV13 [111]	➺ Giving a variant of the RSA threshold scheme in [147] which the simulation proofs of schemes like [143, 266] are difficult to apply since the proofs require the evaluation of non-integer coefficients which may not be able to be evaluated because the denominator is not coprime with modulus ➺ Provable security against static adversary and giving robust version secure against active adversary.
GHKR08 [143]	➺ Proposing an efficient variant of the RSA threshold signature of [266] for a large number of parties in ad-hoc networks and making the protocol independent of the number of parties which had to be fixed and known to all parties ➺ Extending the proposed threshold scheme to support dynamic groups without trusted center using the non-interactive solution [261] ➺ Proving security against a passive adversary in ROM and extending robustness to tolerate malicious adversary.
Sho00 [266]	➺ Proposing a new robust non-interactive RSA threshold signature with provable security against a static adversary in ROM that size of signature share is a multiplication of a constant and RSA modulus ➺ Requiring a trusted center since the RSA modulus is a product of two safe primes ➺ Generalizing the concept of thresholding as t denotes the number of corrupted parties and k denotes the number of shares required for signature generation and proving the security with $k = t + 1$ ➺ Proposing a more general protocol to be used when $k \geq t + 1$ proven in ROM.
Rab98 [251]	➺ Proposing a general paradigm for transforming an all-player protocol into a threshold one ➺ Utilizing the proposed paradigm with RSA signature scheme to obtain interactive threshold RSA ➺ Distributing the RSA secret key as *additive-shares* ➺ Adding robustness by further sharing the share secretly using a secure VSS ➺ Generating signature by multiplying the partial signatures of the parties with possibility of finding malicious party upon facing invalid signature ➺ Achieving proactive RSA by changing the representation of the secret key.

➺ Technical aspects of the scheme ⋆ Additional notes such as performance analysis.

6 Generic Thresholding Approaches

The properties and security models of MPC were discussed in Section 2.4. MPC's inherent functionality makes it well-suited for threshold cryptography, particularly in the context of threshold digital signatures. By employing MPC-based methods, we can achieve *Transparent thresholding*, which allows converting existing signatures into thresholded schemes while preserving their underlying properties. While numerous threshold signature schemes exist across both conventional and PQ settings (see Section 7), many deviate from standardized protocols to reduce the cost of computing complex operations (e.g., hash functions) in MPC (Garbled Circuits). This is common in PQ contexts, where custom designs often diverge from NIST standards. Recent studies [15, 208, 305] show that such deviations, especially the exposure of intermediate values for local efficiency, can introduce vulnerabilities leading to key recovery attacks. Full MPC constructions mitigate these risks by adhering closely to the original signing algorithm, but at the cost of increased communication and computation. In settings such as government systems, where strict standard compliance is essential, non-standard approaches are generally unsuitable. Moreover, a key advantage of standard-compliant threshold schemes is functional interchangeability, which allows for seamless integration with existing verification algorithms.

In the following, we will explore the application of MPC in thresholding conventional signatures. Furthermore, in Section 7, we will investigate the application of MPC in thresholding the NIST's standardized PQ digital signatures.

6.1 Thresholded MPC Protocols

MPC encompasses a range of protocols, and in most modern actively secure protocols, the generation of random data or utilizing private information is often required to be secretly shared. Traditional methods rely on ***Verifiable Secret Sharing (VSS)*** in an information-theoretic model, with a corruption threshold of less than $\frac{n}{3}$ parties. However, more recent protocols operate under a *Full threshold* setting, as incorporated by the SPDZ family [98], Tiny-OT family [196, 233], SPDZ-2k [88], and "n-party GC" family [163, 279] where up to $n - 1$ parties can be corrupted. Specific protocols like "Special GC" [225] achieve a threshold of $(t = 1, n = 3)$, while "General Q2" [269] employs a Q_2 access structure with $t < \frac{n}{2}$ to ensure that no combination of two unqualified subsets can reconstruct the complete set.

Table 8. Summary of MPC-based Threshold Digital Signatures

Work by	Scheme	Threshold	Security
Lin22 [203]	Schnorr, EdDSA	(t, n)	Standard real/ideal paradigm, Proactive adversarial model, UC-security, Identifiable abort
Mei22 [219]	Schnorr	(t, n)	MPC with commitment, UC-security, Identifiable abort
GKMN21 [139]	Schnorr	(t, n)	Dishonest majority, UC Commitments, ZKGCs
BST21 [50]	EdDSA	Q_2 access structure ($t < n/2$)	Active adversarial model
DOKSS20 [93]	ECDSA	$(t, n = 3)$	Security of the underlying ABB and assumption of ECDSA to be secure

6.2 Transparent Thresholding of Conventional Digital Signatures

In the upcoming subsections, we will explore the transparent thresholding of some standard digital signatures using MPC. Table 8 provides an overview of these schemes.

6.2.1 MPC Schnorr. The Schnorr signature and its derivatives (e.g., ECDSA and EdDSA) use confidential elements that support linear combinations. This property enables the creation of multi-party signature variants by applying linear secret sharing methods to these elements. Generating a fresh random value (nonce) for each signature in the original Schnorr signature presents a challenge. Normally, this value is chosen randomly from a specific set ($\mathbb{Z}_q^*$). However, there is no guarantee of a consistently reliable source of entropy, unlike EdDSA, which deterministically calculates the nonce based on the secret key and the message. One solution to tackle the entropy problem is to employ *State continuity*. This involves designing a randomized protocol and utilizing a block cipher with a fresh counter to derive a new random value. However, Parno et al. [243] have demonstrated that even highly isolated devices cannot be fully trusted to maintain the state accurately due to software errors, power interruptions, or malicious attacks.

Garillot et al. [139] proposed a threshold Schnorr signature scheme utilizing stateless nonce derivation, enabled by a zero-knowledge proof system based on the ***Zero-Knowledge from Garbled Circuits (ZKGCs)*** paradigm [172]. This construction eliminates long-term state and remains efficient, requiring at most three AES invocations per AND gate. Two primary challenges arise. First, handling $O(q)$ oblivious transfers and public key operations for q-bit inputs was addressed via a variant of *Committed Oblivious Transfer*, which allows one-time witness commitment and multiple proofs, yielding an 11.5× speedup over standard OT, albeit with a 5× communication overhead. Second, the high cost of translating EC statements into Boolean circuits [75] was mitigated using a garbled exponentiation gadget, achieving a 128× reduction in communication and requiring 62× fewer cipher calls. The protocol deterministically derives signing material by sampling a PRF during key generation and applying it to the message. However, this deterministic approach is not secure against active adversaries. To achieve malicious security, parties must commit to initial randomness and prove the correctness of nonce derivation relative to these commitments using the GMW protocol. This enables secure threshold Schnorr signing without assuming an honest majority, where each party proves that their nonce was correctly computed from the PRF and corresponds to a known discrete logarithm.

6.2.2 MPC ECDSA. Dalskov et al. [93] first proposed a generic transformation from MPC protocols over a field $\mathbb{Z}_p$ to protocols over an EC of order p. This was done via introducing a new ***Arithmetic black-box (ABB)*** functionality where the commands defined in this functionality are assumed to be supported by the MPC. Furthermore, this ABB was extended to support secure computation over an arbitrary abelian group of order p to perform MPC in subgroup G. Extended ABB is sufficient to give a protocol over a group, properties through which linearity is preserved, secret reconstruction is relied only on the group operations in $\mathbb{Z}_p$, and multiplication of secret with public point is possible. Then, they used MP-SPDZ [182] to enable threshold ECDSA signing within different threat models by showing that the MAC scheme of SPDZ can be used to provide share authentication.

6.2.3 MPC EdDSA. MPC enables bit-wise operations on secret values over finite fields, but SS often incurs high communication and complex preprocessing. Keller et al. [183] demonstrated actively secure multi-party Boolean circuit garbling under a dishonest majority, though such approaches remain computationally costly due to modulo reductions or reliance on inefficient two-party protocols. To address these limitations, mixed-protocol frameworks have emerged, combining SS-based MPC for arithmetic operations and GC-based MPC with lower round complexity, with conversion between each mode enabled by ***double-shared authenticated bits (daBits)*** [255].

The use of hash functions in digital signatures presents challenges for thresholding, as it may disrupt the linearity of secret values essential for MPC. EdDSA exemplifies this issue, deriving the ephemeral key by hashing the secret key and message. Two variants exist: the standard variant computes $r = H(H(sk) \| m)$, while *HashEdDSA* uses $r = H(H(sk) \| H(m))$. The latter is more amenable to threshold implementations, as it reduces the MPC complexity by minimizing the hash input size. Bonte et al. [50] showed that threshold EdDSA is feasible with slight adaptations to existing MPC techniques. Building on [268], they use MPC over $\mathbb{F}_q$ for EC operations, with a randomly sampled ephemeral key. For computing the hash functions, two methods are considered: a garbled-circuit-based approach using a variant of the HSS protocol [163] to produce additive authenticated bit-wise sharings; and an SS-based method following Araki et al. [10], which operates over $\mathbb{F}_2$. The latter employs daBits to convert bit-wise sharings into $\mathbb{F}_q$-compatible shares, enabling use of Q_2 access structure and reducing complexity compared to full-threshold models.

7 PQ Secure Threshold Digital Signatures

In 2017, NIST [234] initiated efforts to standardize secure digital signatures that are resistant to quantum attacks. After three rounds of evaluation [235], four schemes were selected for standardization. ML-KEM (CRYSTALS-Kyber) [252] was chosen for PKE and key establishment, while ML-DSA (CRYSTALS-Dilithium) [238], FALCON [135], and SLH-DSA (SPHINCS+) [237] were selected for digital signatures. Moreover, other organizations have also made notable contributions to the advancement of PQC, such as the ***Internet Engineering Task Force (IETF)*** focusing on **Transport Layer Security** (**TLS**) and **Internet Key Exchange** (**IKE**), the ***European Telecommunications Standards Institute (ETSI)*** concentrating on **Quantum Key Distribution** (**QKD**) [294] and PQC with the key exchange standard [126], and the ***International Standards Organization (ISO)*** discussing PQ algorithms in SD8 of JTC 1/SC 27. China, alongside the United States and Europe, has actively engaged in the standardization of PQ schemes. The LAC scheme [209], accepted in a competition organized by the *Chinese Association for Cryptographic Research*, reached the second round of the NIST's standardization. To enable thresholding in PQ-secure digital signatures, a combination of MPC techniques like ***Linear Shamir's secret sharing (LSSS)***-based MPC and *Garbled circuit*-based MPC is required.

After announcing the candidates for standardization, NIST declared the continuation of the PQC standardization process with a fourth round [1]. To preserve diversity in standardized signatures, NIST primarily solicits additional PQ-secure digital signatures based on structures other than lattices. This call aims to benefit applications requiring short signatures and fast verification processes. While the submission of lattice-based signatures, such as HAETAE [81] and Raccoon [101] inspired by Dilithium, and SQUIRRELS [125] inspired by FALCON, is permitted, such submissions are required to outperform previous schemes or include additional security features.

In the following sections, our focus will primarily be on the lattice-based and hash-based approaches employed in the standardized schemes. This study does not cover other approaches, such as isogeny or code-based approaches, which are the basis for non-standardized schemes in previous evaluation rounds.

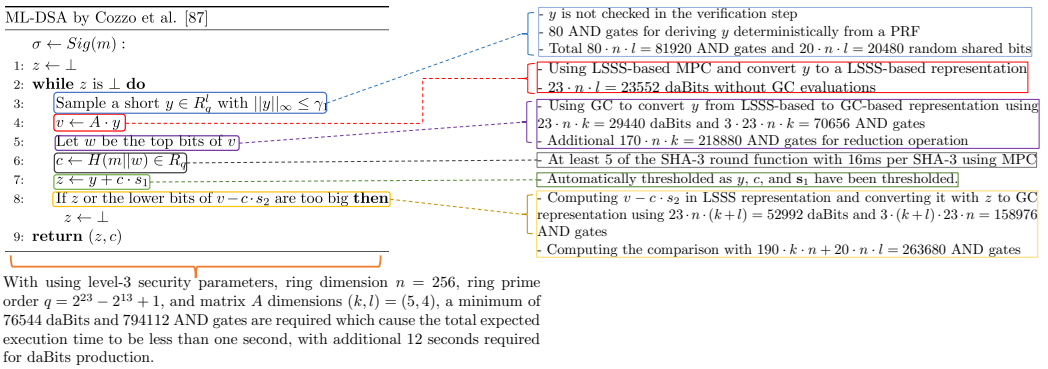


Fig. 3. Threshold ML-DSA (template) digital signature (level-3 security parameters).

7.1 Threshold Lattice-based Digital Signatures

Lattice-based digital signatures have been the focus of extensive research since their introduction by Ajtai [5]. However, many proposed schemes suffer from security vulnerabilities due to private key leakage through signatures. To overcome this, Lyubashevsky et al. [211] introduced the *Fiat-Shamir-with-aborts* methodology to address key leakage in lattice-based schemes. Moreover, rejection sampling is a significant challenge in lattice-based schemes, where intermediate values must be kept confidential until sampling is complete. This requires both GC-based MPC for non-linear operations and LSSS-based MPC for linear operations, along with the required transformation using daBits. These factors pose challenges in achieving thresholding in lattice-based schemes. Lattice-based signature construction, despite its inherent challenges, has seen notable progress in recent years. In particular, 2024 witnessed a surge of contributions [124, 160, 289], including works by Chairattana-Apirom et al. [73], TRaccoon [103], Tang et al. [272], and Ringtail [52], which explored Fiat-Shamir-based designs and MPC-friendly adaptations focused on efficiency, robustness, and active security. This momentum continued with a series of articles by Pino et al. [102, 104, 105], offering scalable and practical threshold signing with improved communication and performance. A comprehensive summary is presented in Table 9.

ML-DSA*:* ML-DSA (formerly known as Dilithium) is a PQ-secure digital signature that incorporates *Multiplication in Polynomial Rings* and *XOF (Extendable-output Function) Expansion* as building blocks. This signature is based on the MLWE problem (See Definition 2.1.8) and follows the Fiat-Shamir-with-aborts paradigm, allowing for convenient adjustment of the security level. Unlike previous lattice-based schemes [115, 116] that used *Gaussian sampling* and were vulnerable to side-channel attacks, this scheme employs uniform sampling with constant-time operations to improve its security and reduce the risk of side-channel attacks. Additionally, it achieves a more efficient design by avoiding Gaussian sampling, reducing overhead for transmitting public keys and signatures. A simplified depiction of the protocol by Cozzo et al. [87] is shown in Figure 3, which includes the thresholding cost of each step and the entire scheme. While existing threshold ML-DSA [37, 51, 117] are functional interchangeable, they do not fully comply with NIST standard (FIPS 204).

FALCON: FALCON is a compact PQ-secure digital signature. Unlike ML-DSA, FALCON adopts the *hash-and-sign* paradigm over Fiat-Shamir to benefit from the GPV framework [146], which provides security in both classical and quantum oracle models, as well as message recovery capability [247]. NTRU lattices within the GPV framework allow for compact instantiation and accelerated operations due to their ring structure. A simplified depiction of the protocol proposed by Cozzo et al. [87] can be

Table 9. Summary of Threshold Lattice-based Digital Signatures

Work by	Description
BCEPT25 [37]	➺ Proposes the first efficient and scalable threshold signing protocol for ML-DSA with signatures fully compatible with the standard verifier ➺ Adapts the ML-DSA signing algorithm for MPC by isolating costly operations (e.g., SHA3) outside the MPC circuit, preserving original security guarantees and improving performance ➺ Operates under an honest-majority setting without additional algebraic assumptions ⋆ Key and signature sizes remain ML-DSA-compliant ⋆ MPC-unfriendly tasks handled locally ⋆ Scales to arbitrary party counts.
DKLS25 [117]	➺ Introduces the first actively secure two-party threshold ML-DSA scheme, *Trilithium*, featuring DKG and functional interoperability with existing verifiers ➺ Assumes an asymmetric threat model: one party is fully malicious, the other is semi-honest (active privacy only), with UC-based security proofs ⋆ Rust implementation achieves signing times of 245, 310, and 364 ms for ML-DSA security levels 2, 3, and 5, respectively.
BCPE$^+$25 [51]	➺ Introducing the first functional interchangeable and practical threshold ML-DSA by presenting two constructions: Threshold Raccoon [103] based scheme supporting up to 64 parties, with identifiable aborts and efficient DKG ML-DSA-compatible threshold scheme supporting up to 6 parties ➺ Utilizes short secret-sharing schemes tailored for lattice settings, enabling compact key shares and low communication overhead ➺ Introduces critical optimizations to support rejection sampling and maintain ML-DSA's norm constraints ⋆ Signatures and verification keys remain identical to ML-DSA ⋆ Protocol messages computed in milliseconds, communication ranges from 10.5 KB to 525 KB per party. Benchmarking in realistic LAN/WAN settings demonstrates practical deployment readiness.
ASY22 [4]	➺ Optimal improving the existing threshold scheme proposed by [45] in various ways: 1. Reducing noise flooding ratio from exponential to sub-linear complexity and hence, increasing the efficiency 2. Addressing the gap in clarification of the underlying signature scheme using a variant of Lyubashevsky's signature [212] that does not rely on rejection sampling 3. Achieving *Partial adaptivity* in the ROM and *Full adaptivity* in the standard model with having preprocessing ➺ Giving a general (t, n) access structure using FHE, UF-CMA signature scheme, and {0, 1}-LSSS.
PET21 [246]	➺ Improving their previous lattice-based **threshold multi-stage secret sharing** (**TMSSS**) scheme [245] such that less public values are published ➺ Proposing a new anonymous and verifiable threshold signature scheme based on the trapdoor function of the [224] and the improved TMSSS where a combiner, leveraging additive homomorphic property of TMSSS, combines partial signatures of the parties and generates a signature ⋆ Improving the share to secret size ratio of the best existing scheme by a factor of two.
VSW21 [276]	➺ Proposing a three-round lattice-based two-party $(2, 2)$ signature scheme constructed from a (bit decomposition)-free variant of the Crystals-Dilithium PQ signature scheme [185] that follows Fiat–Shamir with Abort paradigm with security based on M-LWE and M-SIS (See Definitions 2.1.8 and 2.1.9) proven in the random oracle model ➺ Using uniform distribution instead of discrete Gaussian distribution due to ease of secure implementation ➺ Using homomorphic hash function instead of homomorphic commitments for efficiency ⋆ Significant improvement in the size of the data communicated in the rounds ⋆ No signing comparison due to lack of implementation of other schemes.
BGGJ$^+$18 [45]	➺ Constructing a **threshold FHE** (**TFHE**) with a trusted party based on LWE using two methods of LSSS and plain SSS while managing the noise blow-up ➺ Constructing a three-procedure universal thresholdizer based on the proposed TFHE for thresholding cryptographic primitives including PKE, PRFs, and signatures schemes allowing single-round threshold signature and single-round threshold CCA-secure PKE schemes ➺ Constructing a five-procedure threshold signature and proving the unforgeability, robustness, and compactness ➺ Proving security using different hybrid games ⋆ Compact ciphertext with key share size overhead for {0, 1}-LSSS ⋆ Increasing modulus overhead with more compact key shares for SSS.
WCG15 [278]	➺ Constructing a lattice-based **threshold attribute-based** (**TABE**) signature scheme based on SIS ➺ Incorporating a signature constructed from a proof of knowledge protocol using *Fiat-Shamir heuristic* [133] ➺ Proven selective-predicate and adaptive-message unforgeability in ROM ⋆ Improved private key storage cost with comparable public key and signature sizes.
CK14 [83]	➺ Relying on *hash-and-sign* signature scheme and proposing ***Message Block Sharing (MBS)*** as a tool to construct a (t, n) threshold signature scheme by which a large message is divided into randomized size portions with further giving to the parties using a combinatorial way ➺ Constructing a $(2, 3)$ MBS and further generalizing it to (t, n) ➺ Incorporating the MSB into the lattice-based group signature scheme of [153] to build a threshold signature scheme ⋆ Possible inefficiency due to generating group size after receiving message for signing.
BKP13 [31]	➺ Constructing threshold protocols with three goals: Generating a hard lattice with a strong trapdoor, using the trapdoor to sample from a non-interactive/interactive discrete Gaussian distribution over a desired coset of the lattice, and eliminating the trapdoor for a higher-dimensional extension of the lattice ➺ Having information-theoretic security against an adaptive adversary in the UC framework ➺ Incorporating the proposed protocols as a thresholdizer to construct a lattice-based threshold GPV signature [146].

➺ Technical aspects of the scheme ⋆ Additional notes such as performance analysis.

found in Figure 4, along with the thresholding cost of each step and the entire scheme. The algorithm utilizes the FFT procedure to compute the Fast Fourier Transform of a given polynomial. It employs matrix $\mathbf{B}$ to store randomly generated polynomials during key generation. The $ffSampling$ applies randomized rounding on the coefficients of t and the given FALCON tree T.

7.2 Threshold Hash-based Digital Signatures

SLH-DSA (formerly known as SPHINCS+ [12]) is a stateless hash-based PQ digital signature that builds upon SPHINCS [35] scheme, which incorporates various components of XMSS [58], the first PQ signature scheme. SLH-DSA authenticates a significant number ***Few-time Signatures (FTS)*** using a hypertree structure composed of ***Many-time Signatures (MTS)***, each consisting of a Merkle-tree signature encompassing a ***One-time Signatures (OTS)*** with an authentication path. In terms of keys, the public key is a single hash value derived from the top-level MTS, while the private key is a secret seed value used to generate all FTS and OTS keys.

Based on the performance analysis of SLH-DSA given by [87], the total number of SHA-3 calls in SLH-DSA is 321553. With efficient garbled circuit implementations, the execution time for these calls in the thresholded scheme is estimated to be around 85 minutes. However, following the given order of operations could significantly prolong the execution time by several hundred minutes.

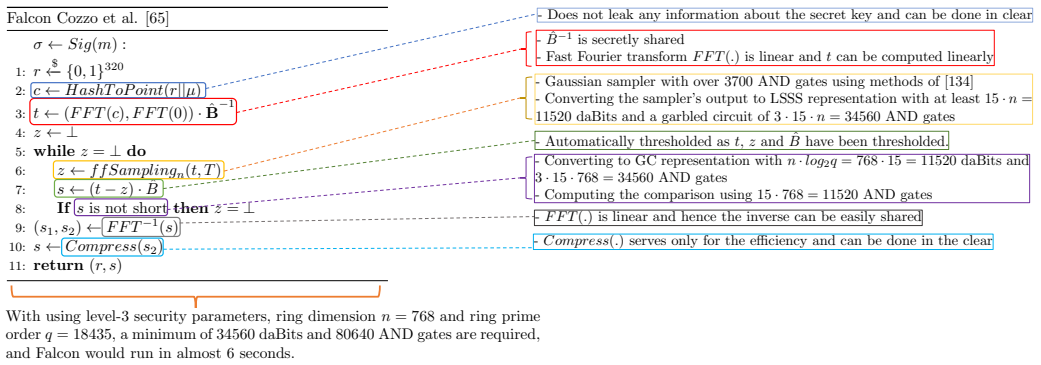


Fig. 4. Threshold FALCON digital signature (level-3 security parameters).

8 Exotic (Special) Digital Schemes

In traditional digital signatures, a single signer signs a message, and one or more recipients verify the signature. In threshold digital signatures, private data (e.g., the private key) is secretly shared among the signers, and a subset of the parties collaborates to generate the signature while the verification process remains the same. In the following, we will explore signatures other than the standards with special properties, specifically Group, Ring, and Multi-signatures.

8.1 Group Signatures

Group signatures involve a group of parties, each possessing their own signing key, who can generate digital signatures on behalf of the group while maintaining a certain level of *Anonymity* and *Traceability*. The signatures are verifiable using the group's single public key by any group member. If the group members are fixed, the group signature is *Static*, and if it allows a user to join and leave, it is *Dynamic*. Setting up and management of the group are all managed by a trusted party known as *Group manager*. Introduced by Chaum and Van Heyst [76], Group signatures initially relied on IF and DLP, with schemes supporting either fixed membership or subgroup formation. Building on this foundation, Camenisch [61] advanced this by enabling dynamic membership with linear-size components. Further advancements were made by Ateniese et al. [11], who formalized key security properties such as *collusion resistance*, *exculpability*, and protection against framing. Bresson et al. [57] addressed revocation for multiple users, while Bellare et al. [27] proposed a formal model incorporating *full anonymity* and *traceability*. Expanding revocation efficiency, Boneh et al. [49] introduced **verifier-local revocation (VLR)** to offload revocation from signers. Kiayias et al. [184] improved scalability using BB signatures [40] and verifiable encryption [63] to enable efficient member joins. In addition to traditional public-key approaches to construct group signatures, recent work by Chen et al. [77] explores the use of MPC-in-the-head non-interactive zero-knowledge proofs. They introduce a novel hash-based group credential scheme, grounded in SLH-DSA, which offers a PQ security alternative to conventional number-theoretic constructions.

With all the foundations of group signatures and improvements made so far, certain studies have concentrated on constructing and improving threshold group signatures [47, 62, 159] in which only a subset (threshold) of the group members sign on behalf of the group. Moreover, recent works are mainly trying to utilize lattices to achieve PQ resistant group signatures, such as works [64, 67] to obtain static group signatures, [300, 301] to create group signatures with VLR, [178, 258] to achieve partially-dynamic and [178, 270] to achieve fully-dynamic group signatures.

8.2 Ring Signatures

Ring signatures enable the creation of digital signatures without setting up procedures, group managers, or revocation mechanisms and offer signer anonymity within a defined group of potential signers. In these schemes, each participant possesses a private and public key. When signing a message, each party selects a subset of public keys, including their own, to form an anonymous group known as a *ring*. This ring is transmitted alongside the signature. During verification, the recipient uses the ring to establish that one of the ring members generated the signature.

Ring signatures were first introduced by Rivest et al. [254] using RSA, initiating formal treatment of signer anonymity within a group. Abe et al. [2] soon proposed a DLP-based variant, while Naor [230] introduced undeniable ring signatures by combining encryption [254] with deniable authentication [118], later extending them to threshold settings. Boneh et al. [46] leveraged aggregate signatures from Gap groups to construct efficient ring signatures. To eliminate trusted setup, Dodis et al. [107] introduced the AHAI primitive, achieving constant-size output and computation. Threshold ring signatures were explored in [14, 227, 239], while Esgin et al. [121] introduced the first practical lattice-based scheme, further optimized in their later work [122]. Scalable constructions followed [36, 213], with MatRiCT [123] maintaining efficiency across varying ring sizes. Yuen et al. [298] proposed linear-size signatures, and Katz et al. [181] introduced a hash-based design using NIZKPoKs. Recently, Chiang et al. [82] proposed a PQ-secure threshold ring signature via VOLE-in-the-head, and Duan et al. [114] designed a linkable threshold variant for confidential transactions.

8.3 Multi-signatures

Multi-signatures involve a subset of players collaborating to create a signature, while each member in the subgroup has their own private and public key pair. These schemes allow the verifier to confirm the participation of all subgroup members. Unlike threshold digital signatures that require a predefined subgroup size, multi-signatures have no limitations on subgroup size. Unlike group signatures and ring signatures, where a single signer can generate a signature on behalf of the entire group or ring, these require the participation of all the members, resembling a full threshold signature.

Multi-signatures were first introduced by Itakura et al. [170], with Boldyreva [39] later proposing a BLS-based scheme that avoided fixed subgroups and maintained constant signature size and verification cost, independent of the number of signers. Unlike the approach of Micali et al. [223], Boldyreva's construction supported concurrent executions by eliminating the need for rewinding. In 2019, Maxwell et al. [218] introduced MuSig, a three-round Schnorr-based protocol that enables key aggregation and is secure in the plain model. To address issues with nonce reuse [248], Nick et al. [232] proposed MuSig-DN, the first deterministic Schnorr multi-signature, using PRF-based nonce generation. This was followed by MuSig2 [231], which reduced the protocol to two rounds while preserving security, efficiency, and support for concurrent signing. Recent works have focused on PQ-secure multi-signatures, particularly under MLWE and MSIS assumptions (see Definitions 2.1.8 and 2.1.9). El Kaafarani et al. [119] introduced a three-round construction based on GLP [157], using Ring-LWE and Ring-SIS. Damgard et al. [97] addressed security gaps in this scheme, offering two improved variants at the cost of efficiency. Ma et al. [214] achieved shorter signatures, while Boschini et al. [53] proposed a practical trapdoor-free lattice-based alternative, avoiding the complexity of [97].

9 Potential Applications for Threshold Signatures

Threshold digital signatures, especially those with exotic properties like group and ring, are employed across diverse domains such as blockchain, cryptocurrency, IoT, cloud computing, and

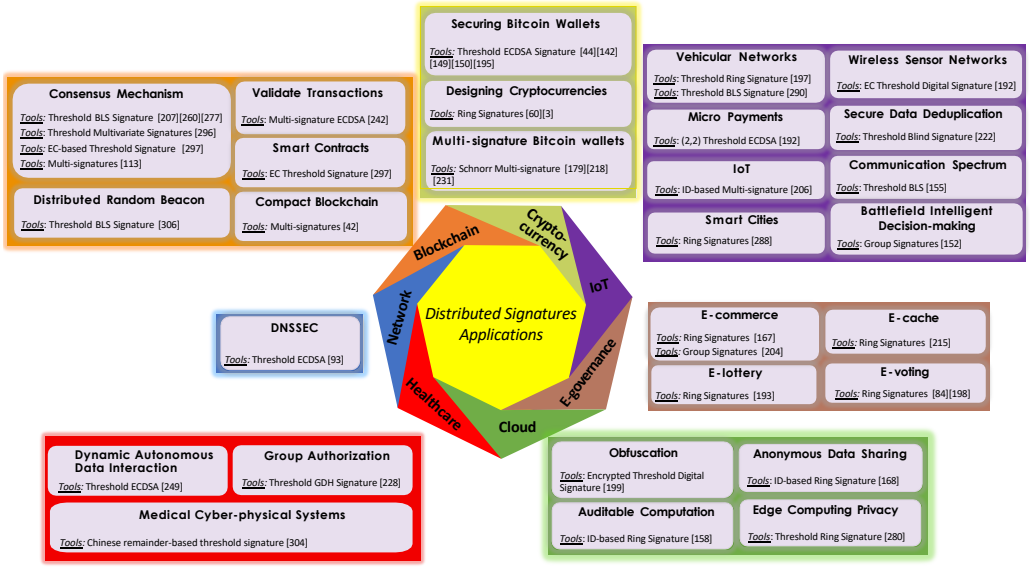


Fig. 5. Applications of distributed digital signatures.

healthcare. Figure 5 illustrates a taxonomy of these applications. Below, we highlight representative use cases from each domain.

Blockchain and Cryptocurrency: Threshold and multi-signatures enhance decentralized consensus, transaction validation, and smart contract execution. Applications include multi-signatures [113], threshold BLS [207, 260, 277], and other threshold schemes [296, 297]. Threshold BLS also supports distributed randomness via random beacons [306], while compact blockchain designs benefit from aggregated signatures [42]. ECDSA-based multi-signatures are used for efficient transaction validation [242]. In Cryptocurrency, threshold ECDSA has been widely adopted to secure Bitcoin wallets [44, 142, 149, 195, 229]. Schnorr-based multi-signatures [179, 218, 231] enable efficient wallet constructions with key aggregation. For anonymity purposes, cryptocurrencies like Bytecoin [60] and Aeon [3] utilize ring signatures.

IoT: Distributed signatures address security and privacy in IoT systems. Linkable ring signatures support vehicular networks and smart cities [197, 288], while threshold ECDSA [192] and BLS [155, 290] enable secure microtransactions and network coordination. Other use cases include identity-based ring signatures [206], threshold blind signatures for deduplication [222], and military systems [152].

Electronic Governance: Privacy-preserving primitives such as ring signatures are used in e-voting [84, 198], e-cash [215], and e-lottery [193]. Group and ring signatures also support secure e-commerce [167].

Cloud Computing and Network Security: Ring and threshold signatures ensure secure and anonymous cloud interactions. ID-based ring signatures support data sharing and auditing [158, 168], while threshold ring signatures enhance edge computing privacy [280]. Encrypted threshold signatures enable secure obfuscation [199]. Threshold ECDSA strengthens DNSSEC by protecting private keys during domain authentication, as shown by Dalskov et al. [93].

Healthcare: Threshold group signatures support dynamic data access in healthcare systems [249] and threshold DH-based and CRT-based signatures enable group authorization [228] in medical cyber-physical systems [304].

10 Open Problems, Challenges, and Future Directions

Despite substantial advancements in threshold signature schemes, spanning both conventional primitives (e.g., Schnorr, ECDSA) and PQ constructions (e.g., ML-DSA), several critical challenges and open research questions persist.

Side-Channel Attacks in Threshold Signature Implementations: While side-channel attacks are well studied in single-signer cryptography, their impact on distributed and threshold signatures remains underexplored. Kyster et al. [194] identified a MAC key leakage in the SPDZ MAC-check protocol due to concurrent multi-threaded execution, affecting output integrity and input privacy. Given similar concurrency in threshold schemes, this raises serious concerns about their side-channel resilience, potentially requiring significant software or architectural mitigation.

Security Challenges in Thresholding ML-DSA: Efficiency vs. Leakage Tradeoffs: Recent threshold constructions for lattice-based signatures, particularly ML-DSA, vary from closely following the original design to introducing structural optimizations with formal security proofs. A common efficiency technique is local share reconstruction, allowing parties to perform expensive computations (e.g., extendable-output functions) off-chain without using MPC (e.g., Garbled Circuit). While this reduces overhead, it introduces largely unexplored risks of private key leakage. Recent works [15, 208, 305] demonstrate that exposing both the challenge and response (as a result of performance improvement techniques) in rejected signing attempts can enable secret key recovery, challenging the previous threshold ML-DSA constructions.

Underutilization of Threshold Signatures in Diverse Domains: As noted in the Applications section, research on threshold signatures has predominantly focused on blockchain-related use cases, while other domains, (e.g., healthcare and PKI key compromise) remain relatively underexplored. Expanding the applications could reveal new challenges and requirements, potentially exposing limitations in current schemes and driving further innovation in threshold cryptography.

Regulatory and Standardization Constraints on Custom Threshold Implementations: Custom threshold designs often balance efficiency and security but face limited adoption in regulated settings due to misalignment with established standards, particularly those from NIST. Many diverge from standardized constructions, exposing them to emerging attacks [15, 208, 305]. An exception is Cozzo et al. [87], who provide one of the few cost estimates for thresholding NIST PQ-signatures using pure MPC while preserving standard compliance. However, practical implementations and detailed cost analyses of MPC-based schemes remain scarce, hindering their evaluation in compliance-driven environments.

Balancing Privacy and Accountability in Threshold Signatures: Previous threshold signature constructions have focused on two distinct approaches: one prioritizing signer anonymity and the other ensuring accountability for identifying signers. Applications such as blockchain require a balance between these two factors. This need led to the development of the **Threshold, Accountable, and Private Signatures** (**TAPS**) scheme [47]. Despite limited research on TAPS, there is significant potential for developing implementations based on standard assumptions, offering advantages such as shorter public keys, compact signatures, full traceability, and efficient verification.

11 Conclusion

This article presents a comprehensive and systematic overview of threshold and exotic digital signatures, highlighting their pivotal role in ensuring security within the evolving landscape of decentralized NextG networked systems and applications. We first begin by examining existing literature and identifying limitations and gaps that need to be addressed. Then, our analysis extends to both conventional and PQ cryptography scenarios, including custom-designed and standard signatures like NIST and NIST-PQC. In this analysis, for both conventional and PQ-secure signatures, we consider the generic methods, such as secure MPC, as well as specific thresholding techniques

applied across diverse signature families. Finally, we examine the application of these signatures in real-life contexts and provide a forward-looking perspective for future research.

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